Uncooperative Spacecraft Pose Estimation using Monocular Monochromatic Images

Jian-Feng Shi [a], Steve Ulrich [*]

Imaging cameras are cost-effective sensors for spacecraft navigation. Image-driven techniques to extract the target spacecraft from its background are efficient and do not require pre-training. In this paper, We introduce several image-driven foreground extraction methods, including combining the difference of Gaussian-based scene detection, and graph manifold ranking based foreground saliency generation. We successfully apply our foreground extraction method on infrared images from the STS-135 flight mission captured by the Space Shuttle's *TriDAR* thermal camera. Our saliency approach demonstrates state-of-the-art performance and provides an order of magnitude reduction in processing speed from the traditional methods. Furthermore, we develop a new uncooperative spacecraft pose estimation method by combining our foreground extraction technique with the level-set region-based pose estimation with novel initialisation and gradient descent enhancements. Our method is validated using synthetically generated Envisat, Radarsat model, and International Space Station motion sequences. The proposed process is also validated with real rendezvous flight images of the International Space Station.

Nomenclature

$\mathbf{0}, \mathbf{1}$	=	zero and identity matrix respectively
$\nabla(\mathbf{x})$	=	gradient function
α	=	graph manifold ranking scaling parameter
$\boldsymbol{\epsilon}, \eta$	=	$[x\ y\ z]^T$, w quaternion respectively
Ω_i	=	image domain where i is foreground (f) or background (b)
Φ	=	level-set embedding function

[a]Senior GNC Engineer, MDA, 9445 Airport Road, Brampton, Ontario L6S 4J3, Canada, Senior Member AIAA.
[*]Associate Professor, Department of Mechanical and Aerospace Engineering, Carleton University, 1125 Colonel By Drive, Ottawa, Ontario K1S 5B6, Canada, Senior Member AIAA.

μ_i, σ_i	=	pixel mean and standard deviation where i is foreground (f) or background (b)
$\tilde{\mathbf{A}}$	=	unnormalised optimal affinity matrix
$\mathbf{A}, \mathbf{B}, \mathbf{C}$	=	pose quaternion matrices
$\mathcal{B}(\mathbf{x})$	=	box filter operator
C	=	image curve entity
$\mathcal{CG}(\mathbf{x}), \mathcal{CL}(\mathbf{x})$	=	image converting functions from colour to grey and *RGB* to *CIE Lab* respectively
$\mathbf{D}$	=	degree matrix
$\hat{D}, \check{D}$	=	maximum and minimum feature vector difference respectively
$d^{(ij)}$	=	i^{th} row, j^{th} column element in the degree matrix
$\overline{\mathcal{D}}(\mathbf{x})$	=	normlised difference of Gaussian function
E	=	image energy function
f	=	camera focal length
$\mathbf{f}, \mathbf{f}^*$	=	superpixel and optimised graph manifold ranking function respectively
$\mathcal{F}(\mathbf{x})$	=	fast explicit diffusion function
$\overrightarrow{\mathcal{F}}$	=	coordinate frame vectrix
$\hat{\mathbf{f}}_i$	=	unit vector of the i vector gradient
$\mathbf{G}$	=	11×11 Gaussian filter
$\mathcal{G}(\mathbf{x})$	=	Gaussian distribution map generation function
$\mathbf{h}^{(i)}$	=	feature vector for the ith superpixel
h_i	=	stepsize in the i direction where i is x, y
$H(\bar{z}), \delta(\bar{z})$	=	Heaviside-step and smoothed Dirac-delta function respectively
$\mathbf{I}, \bar{\mathbf{I}}$	=	input colour image and normalised image respectively
$\mathbf{I}_{Lab}$	=	image in *CIE Lab*
$\mathbf{K}$	=	intrinsic camera matrix
$\mathbf{K}_B, K_B$	=	box filter kernel and kernel size respectively

$\mathbf{K}_i, K_i$	=	elliptical kernel and kernel size respectively where i is dilation (d) or erosion (e)
K_L	=	Laplace filter kernel size
$\mathbf{K}_s$	=	3×3 Scharr kernel
$\mathcal{L}(\mathbf{x})$	=	Laplacian filter
M_i	=	appearance model where i is foreground (f) or background (b)
$\mathcal{M}(\mathbf{x})$	=	mean value binary threshold function
$\mathcal{MB}(\mathbf{x})$	=	minimum boundary distance saliency response generation function
$\mathcal{MC}(\mathbf{x})$	=	image moment centroid generation function
$\mathcal{MD}(\mathbf{x}), \mathcal{ME}(\mathbf{x})$	=	morphological dilation and erosion function respectively
$\mathcal{N}(\mathbf{x})$	=	image nomalisation function
$\mathcal{O}(\mathbf{x})$	=	Otsu thresholding function
o_i	=	image center position in the i direction where i is x, y
P_i	=	probability density function where i is foreground (f) or background (b)
$\mathbf{q}, q_i$	=	quaternion represented target orientation and i^{th} parameter quaternion respectively
$\tilde{\mathbf{q}}$	=	non-normalised quaternion
$\mathbf{R}$	=	rotation from target body frame to camera frame
$\mathcal{R}(\mathbf{x})$	=	apply contours and keep only maximum perimeter region
$\mathcal{RC}(\mathbf{x})$	=	regional contrast saliency response generation function
r_{tol}	=	foreground/background classification ratio tolerance
$r_i(\mathbf{x})$	=	likelihood of the pixel property where i is foreground (f) or background (b)
$\mathbf{S}_{bg}, \mathbf{S}_{fg}$	=	background and foreground mask respectively
$\mathbf{S}_d$	=	simplified DoG normalised response map
$\mathbf{S}_e, \mathbf{S}_{fe}$	=	Scharr filtered edge response map and edge contour response respectively
$\bar{S}_{fed}$	=	mean value FED response map
$\mathbf{S}_G$	=	Gaussian centroid response map

$\mathbf{S}_{hsf}$	=	high-frequency saliency feature response map
S_i	=	camera image scaling in image i direction where i is x, y, θ. θ is the skew scaling
$\mathbf{S}_i$	=	normalised image intersected with i is foreground (f) or background (b) mask
$\mathbf{S}_L$	=	Laplace algorithm foreground response map
$\mathcal{SL}(\mathbf{x})$	=	SLIC superpixel generation function
$\mathbf{S}_m, \mathbf{S}_M$	=	mean value binary threshold contours and response map respectively
$\mathbf{S}_{mask}$	=	processed foreground mask
$\mathbf{S}_{mbd}$	=	minimum boundary distance response map
$\mathbf{S}_u$	=	main region contour mask
$\mathbf{S}_o, \mathbf{S}_O$	=	contour region Otsu threshold and Otsu thresholded response map respectively
$\mathbf{S}_p$	=	saliency enhanced response map
$\mathbf{S}_{rc}$	=	regional contrast response map
$\mathcal{SR}(\mathbf{x}), \mathbf{S}_{SR}$	=	spectrial residual response function and response map respectively
$\mathbf{t}$	=	translation vector from camera to target body expressed in camera frame
$\tilde{\mathbf{t}}$	=	non-centralised initial translation vector
$\mathbf{T}_i$	=	transformation matrix for the i axis where i is x, y, z
$\mathbf{W}, w^{(ij)}$	=	affinity matrix and i^{th} row, j^{th} column parameter in the affinity matrix
$\bar{x}, \bar{y}$	=	x,y-pixel location from image center
x, y, z	=	translational x,y,z or for x,y used as image pixel position
X, Y, Z	=	x,y,z-axis direction
$\mathbf{x}, \mathbf{x}_k$	=	input image or pose state vector at the kth step
$\mathbf{X}_i$	=	target body coordinate where i is the body (b) or camera (c) frame
$\mathbf{x}_{\gamma_i}$	=	image coordinate partial derivative with respect to γ_i direction
$\mathbf{y}$	=	pixel intensity or colour feature vector
$\mathbf{z}$	=	manifold ranking indication vector

I. Introduction

Vision and LIDAR based spacecraft rendezvous and docking is a topic of interest in the spacecraft Guidance, Navigation, and Control (GNC) community [1–4]. Precise visual pose estimation is typically performed using prepared targets with fiducial markers [5–8]. Other research have used image features to determine unprepared spacecraft pose [9–13]; however, when the visual scene is cluttered with the Earth as background, extracting useful image feature becomes more difficult [14]. Some have used shadow prediction and regions for simple shapes [15, 16]; complex shapes on the other hand, requires innovative ways to extract the foreground from a cluttered background [17].

Spacecraft pose computed from monochromatic cameras is known to be a key requirement for autonomous GNC systems; however, image patterns from the background scene may complicate the pose analysis. While recent techniques in deep learning offer precise foreground extraction [18, 19], they nevertheless require pre-training and quality training data that are time consuming to generate. To this end, recent work by Kisantal *et al*. [20] developed photo-realistic spacecraft pose estimation dataset for machine learning algorithm training and inference. Our approach developed in this paper limits image clutter by extracting spacecraft foreground using saliency detection. Specifically, we use image-driven approach to detect and extract the spacecraft target in real-time and reduce pose estimation processing time and prediction errors.

Object image pose estimation is a core problem in computer vision, Rosenhahn *et al*. [21], and Lepetit and Fua [22] summarised methods using feature-based structures such as points, lines, circles, and chains to free-form objects such as active contours and implicit surfaces. The state-of-the-art methods in spacecraft pose estimation are based on image features and registration. Both Sharma *et al*. [12] and Capuano *et al*. [23] present similar pipelines using region of interest estimation, feature extraction, and point or shape registration. As Capuano *et al*. [23] points out, *the assumption of a number of pixels in the region of interest must be large enough to allow the extraction of a minimum number of point features.* In order for the extracted features to be useful, the region of interest must be estimated precisely. Unsupervised image processing methods [12] for region estimation can be quickly generated, it may, however, be confused by the intense geographical features of the Earth background. Alternatively, regions of interest can be achieved with good confidence using machine learning or deep learning [24, 25] techniques, it is, however, a highly time-consuming task [23]. The difficulty in the latter feature extraction assumption is often caused by intense space lighting and shadowing.

Keypoints based on image corners [26, 27] and extrema image blobs [28, 29] can be difficult to classify due to its quantity and quality under extreme lighting or non-distinctive visual targets. The feature descriptors used for keypoint matching can be time-consuming to compute and can be erroneous especially under affine or viewpoint transformations and illumination distortions. Edge-based pose estimation [30] represents stable features to track with many invariant properties, but extracting relevant edge features and discarding line clutters requires robust decision tools. On the other hand, the region-based pose estimation uses invariant region boundaries with better recognition

properties. In this context, we develop a real-time region-based method for monocular infrared (IR) image spacecraft pose estimation.

II. Related Work

We organise foreground object extraction into three branches of computer vision research: techniques in using sequential images for background subtraction [31, 32], techniques in machine learning or deep learning neural networks for semantic segmentation [33, 34], and image-driven saliency detection [35, 36]. Many background subtraction methods require multiple image sequences to train a model prior to the foreground extraction. Similarly, machine learning methods also demand training a classifier before inference. Particularly in the case of some convolutional neural networks, while hundreds of labelled images can be used to perform transfer learning on an established network, millions of labelled images are required to develop weights in a non-intialised network [37]. Saliency methods, on the other hand, can either be top-down [35], bottom-up [36], or a mixture of both [38]. Top-down methods refer to using training images to establish a foreground feature database then using a classifier to extract specific image regions; this database can be used to retrieve the desired foreground object from the image. The bottom-up approach is image-driven, it is usually faster and easier to implement since it does not require external training as a separate process.

Three recent real-time capable saliency detection methods are Regional Contrast (RC) [39], Minimum Barrier Distance (MBD) [40] and Graph Manifold Ranking (GMR) [41]. Regional Contrast was proposed by Cheng *et al*. [39] to extract saliency from local regions as the weighted sum of colour contrast. Regional contrast can be computed efficiently and can produce precise foreground from colour images. Zhang *et al*. [40] proposed a fast raster-scanning algorithm to approximate the MBD transform [42]. An extended version, MB+, was proposed by adding *Image Boundary Contrast* map using border pixels as colour contrast seeds in the whitened colour space. Finally, Yang *et al*. [41] proposed GMR saliency using document ranking; GMR optimises a graph-based ranking cost function while replacing documents with image superpixels. The aforementioned saliency detection methods have better precision working with coloured images; however, their performance degrades if the image is monochromatic. Greyscale image saliency include using gradient orientation [43] and texture [44]. The first original contribution of this paper is to propose several saliency models; our most effective saliency model uses an enhanced GMR [41] in combination with MBD [40], RC [39], and Difference of Gaussian (DoG)-based background identification. Our method is designed for greyscale images to improve the prior mask used in the level-set regional pose estimation.

Some of the newly developed methods in level-set based segmentation and pose estimation are as follows: Dambreville *et al*. [45] projected 3D models on to the two-dimensional (2D) image and used region-based active contours for shape matching, his method showed robustness to occlusions. Prisacariu and Reid used posterior membership probabilities for foreground and background pixel in support of real-time tracking in the PWP3D model [46]. Perez-Yus *et al*. [47] used a combination of depth and colour for robustness to occlusions in the level-set segmentation and pose estimation. Hexner and Hagege [48] proposed local templates to enhance the PWP3D global framework.

Tjaden and Schömer [49] proposed a pixel-wise optimisation strategy based on a Gaub-Newton-like algorithm using linearised twist for pose parameterisation. Tjaden and Schömer [50] also proposed a temporally consistent, local colour histogram as an object descriptor for the level-set pose estimation template. The general trend for level-set pose estimation research is enhancing model compatibility. We take a different approach to improve the input image to achieve better performance.

To summarise, the main contribution of this paper is the development of novel unsupervised scene recognition and foreground extraction using image saliency principles. We provide a spacecraft image dataset complete with ground truth foreground masks for model experimentation and evaluation. We contribute to the non-cooperative region-based pose estimation technique by combining the level-set segmentation approach with our scene recognition and foreground extraction method. Furthermore, this paper introduces innovative pose initialisation and gradient descent approaches to speed-up and stabilise the pose convergence process.

This paper is organised as follows: Section III provides original saliency model techniques for foreground extraction and scene classification. Section IV provides the enhanced region-based level-set pose estimation method. Section V provides details on benchmark methods, dataset, metrics, computing platform, and test image details. Section VI provides the results and discussions for the proposed saliency detection and pose estimation models. Finally, Sec. VII concludes this work.

III. Image Processing Method

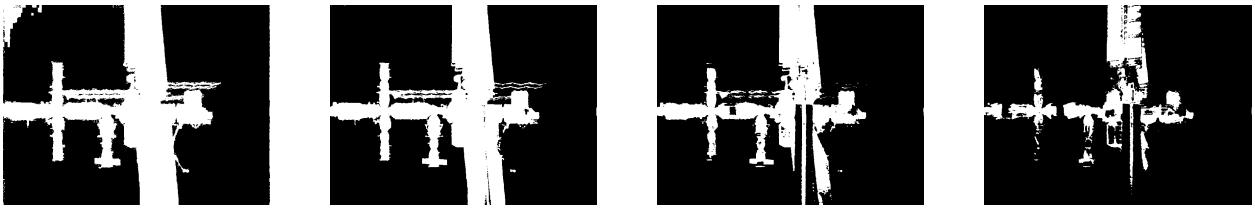
We define two scenarios in the spacecraft rendezvous imagery; the *nadir*, *i.e*. Earth pointing, and *non-nadir* pointing by the chaser spacecraft camera. In the latter case, the target spacecraft can easily be extracted using thresholding since the background is generally black with some minor polluting light source from stars or camera distortions. Strong light source from the Moon or the Sun are localised and rarely occur. The nadir pointing scenario is much more difficult to resolve, especially for monochromatic images. The Earth background can clutter the input image with clouds, land patterns, and brightly reflected sunlight from the oceans. Operationally, a nadir view during rendezvous operation is unavoidable. For example, there is only one zenith-wise corridor for logistical vehicles to approach the International Space Station (ISS); therefore the view of the target spacecraft from the ISS will always have the Earth backdrop. Another example is in a geostationary servicing mission; the most logical docking face is on the anti-Earth deck which houses the launch adaptor ring; therefore the servicing vehicle must approach the target satellite from the nadir direction facing the Earth.

The computation resource required for both scenarios is vastly different; the non-nadir view can be computed using fast adaptive thresholding, whereas the nadir view requires more software intelligence to extract the foreground target. In the first part of this paper, we developed a reliable unsupervised foreground extraction technique for the non-nadir scenario. At the same time, this technique detects possible Earth background so the image processing can be transfer to a more complex algorithm to separate the foreground vehicle from the Earth. It is possible to use the results

from attitude motion tracking in combination with inertial navigation sensors to determine the spacecraft pointing direction. Our approach purely relies on the image sensor to avoid dependencies on spacecraft attitude determination systems. The image only approach provides more flexibility when integrated into the spacecraft pose estimation pipeline. Furthermore, when the camera system is pointing forward or aft, the proposed foreground extraction methods are useful tools for system designers. While we have not explored into sensor fusion and combined systems, they are also valid ways of solving the same background scene classification problem and can be revisited in the future.

A. Scene Classification

During the non-nadir pointing scenario, only the backdrop of black space is behind the target vehicle. One may consider the input image already as a form of the foreground saliency map without any additional processing. A closer examination, however, shows there can be various defects in the image. Figure 1 shows increasing threshold level of an ISS IR image, where values under each figure represent the intensity threshold value. Figure 1(a) shows hardware heating regions can be detected in the upper left corner when the threshold level is 10; this is due to thermal sensor bias that is continuously changing over time. On the other hand, nearly 30 percent of the vehicle is non-visible when the threshold is 100. The threshold level of 60 removes the Space Station Remote Manipulator System (SSRMS) and the lower solar panel. Out of the four images, a threshold of 30 is the closest match to the ground truth while some of the border sensor noise remains visible.



(a) 10 (b) 30 (c) 60 (d) 100

Fig. 1 ISS IR Image Threshold Level Variation Increasing from 10 to 100. Pixel Detectability is Reduced with Increasing Threshold Level.

Our method for region detection and background identification is provided in Algorithm 1 and Fig. 3, where the foreground (FG) and background (BG) histograms are red and blue lines in Fig. 3(l) respectively. First, we resize the input image, $\mathbf{I}$, to 120 rows and normalise the resized image between 0 to 255, $\bar{\mathbf{I}} = \mathcal{N}(\mathbf{I})$. Using the experimental datasets described in Sec. V, 120 rows provide the optimal precision versus speed against 60 and 240 row input image resize. The size reduction of the input image is the main reason for computation speed increase; however, precision will reduce when the reduced image is used by conventional methods. Our formulation ensures the output precision from reduced input image remains equal or better than the conventional methods on the original image with a significant increase in computation speed. Next, we apply mean value binary thresholding, $\mathcal{M}(\mathbf{x})$, on the image to remove low-intensity defects such as image sensor local bias and background starlight. If the detector is operating during the nadir phase, the bright Earth background will remain unaffected. A drawback of thresholding is the creation of holes when

removing shadowed regions; it will also keep dust particles from image noise. We extract contours from the threshold mask and keep only the maximum perimeter region using the operator $\mathcal{R}(\mathbf{x})$. The main region contour will fill in holes and remove all dust particles from it; we find this to be more effective than using an *open* morphological operation. The main region contour mask, $\mathbf{S}_u$, has fine border resolution, but it is only the result of illumination intensity, it will not exclude the Earth background if the camera is nadir pointing.

To distinguish which direction the camera is pointing to, we use the intensity on either side of the high-frequency response to decide the region class. High-frequency edge features are the result of spacecraft boundaries, internal vehicle connections, Earth horizon, and sharp Earth textures such as coasts and other geological boundaries. During the Earth passage, illuminated regions will occur on both sides of the border feature whereas if the imager is non-nadir pointing, the most brightly lit region will only occur on the inside of the outer spacecraft border. We use this crucial observation as the decision rule to classify the pointing direction. To extract the high frequency content, we first approximate the gradient of the normalised input image, $\mathbf{S}_e = \nabla\left(\bar{\mathbf{I}}\right)$, by convoluting $\bar{\mathbf{I}}$ with the 3×3 Scharr kernel [17], $\mathbf{K}_s$. To extract the local region, we increase the edge responses by taking the DoG. The DoG is a faster numerical approximation of the Laplacian of Gaussian from the heat equation, and it will diffuse the edge response uniformly in all directions surrounding the high-frequency edges. The normalised response map, $\mathbf{S}_d$, is a faster, and simplified form of the DoG [28] by convoluting the edge response with the difference of two Gaussian kernels of varying scales. We denote the normalised DoG as $\overline{\mathcal{D}}(\mathbf{x})$.

We also compare using the linear diffusion heat equation with the non-linear diffusion equation. We compute the non-linear diffusion filtering using Fast Explicit Diffusion (FED) [29]. We find similar results in precision using both approaches; the DoG approach takes nearly half the FED computation time using one FED iteration step. Figure 2 shows the DoG and FED response maps and the respective final foreground masks, where $\bar{S}_{fed}$ is the mean value of $\mathcal{F}\left(\bar{\mathbf{I}}, \mathbf{S}_e\right)$, and the FED results are slightly more accurate by preserving the end-effector on the SSRMS. Next, we compute the foreground mask, $\mathbf{S}_{fg}$, by applying both the Otsu [51] and mean binary threshold to $\mathbf{S}_d$ and union with $\mathbf{S}_u$ to fill large gaps and holes. We can compute the background mask, $\mathbf{S}_{bg}$, by taking the bitwise *NOT* operator of the foreground map. The foreground and background pixel maps, $\mathbf{S}_f$ and $\mathbf{S}_b$ respectively, are the intersect of $\bar{\mathbf{I}}$ and the respective masks. After computing the mean and standard deviation of the foreground and background intensity histograms, we formulate our decision rule for the pointing phase classification as

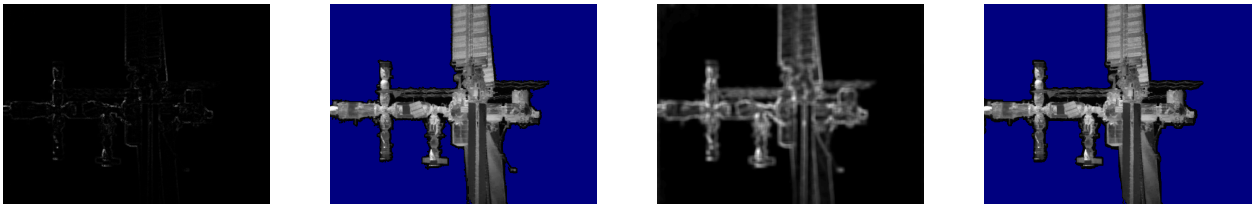
$$\left(\frac{\mu_b + \sigma_b}{\mu_f + \sigma_f} \; < \; r_{tol}\right) = \begin{cases} 1 & \text{non-nadir phase} \\ 0 & \text{nadir phase} \end{cases}, \tag{1}$$

where μ_i and σ_i, $i \in \{f, b\}$, are the mean and standard deviation of the foreground and background intensity histograms respectively. During Earth passage, the background will become brighter and the mean and standard deviation for the background histogram will shift higher. Based on this observation, Eq. (1) separates two highly distinctive pointing

phase described by the intensity histograms, an example is shown in of Fig. 3(l). We select the ratio tolerance to $r_{tol} = 0.2$ in our implementation. If the image class is non-nadir pointing, we confirm the foreground mask by intersecting $\mathbf{S}_{fg}$ with the main region from the edge response, $\mathbf{S}_{fe}$. The benefits of the intersection can be observed in Fig. 4, where the image computed with $\mathbf{S}_{fe}$ has more refined foreground map beneath the right solar panel and around the solar wing connections. Figure 4(a) is the original image, subfigures (b) and (c) are the foreground mask intersection without and with $\mathbf{S}_{fe}$ respectively. If the image class is nadir pointing, more sophisticated saliency detection is needed to extract the foreground spacecraft, refer to Sec. III.B for details. Finally, we resize the foreground mask to the input image size if required. An automatic thresholding technique by Otsu [51] selects the optimal

Algorithm 1 The *REGION_DETECT* Foreground Mask Algorithm

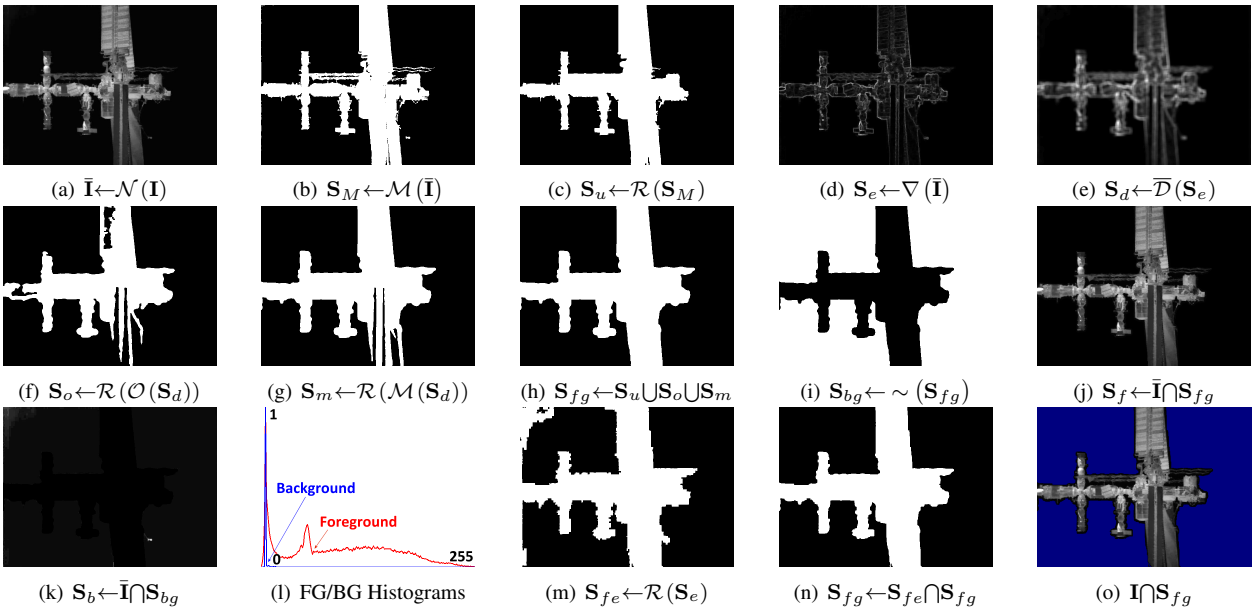
procedure REGION_DETECT($\mathbf{I}$)
  if Image larger than 120 rows **then**
    Resize input image to 120 rows
    Resized = TRUE
  $\bar{\mathbf{I}} \leftarrow \mathcal{N}(\mathbf{I})$
  $\mathbf{S}_u \leftarrow \mathcal{R}\left(\mathcal{M}\left(\bar{\mathbf{I}}\right)\right)$
  $\mathbf{S}_e \leftarrow \mathbf{K}_s^{(3\times3)} * \bar{\mathbf{I}}$
  $\mathbf{S}_d \leftarrow \mathcal{N}\left(\left(\mathbf{G}\left(\sqrt{2}^{7}\right) - \mathbf{G}\left(\sqrt{2}^{-7}\right)\right)^{(11\times11)} * \mathbf{S}_e\right)$
  $\mathbf{S}_o \leftarrow \mathcal{R}\left(\mathcal{O}\left(\mathbf{S}_d\right)\right)$
  $\mathbf{S}_m \leftarrow \mathcal{R}\left(\mathcal{M}\left(\mathbf{S}_d\right)\right)$
  $\mathbf{S}_{fg} \leftarrow \left(\mathbf{S}_u \bigcup \mathbf{S}_o \bigcup \mathbf{S}_m\right)$
  $\mathbf{S}_{bg} \leftarrow \sim\left(\mathbf{S}_{fg}\right)$
  $\mathbf{S}_f \leftarrow \bar{\mathbf{I}} \bigcap \mathbf{S}_{fg}$
  $\mathbf{S}_b \leftarrow \bar{\mathbf{I}} \bigcap \mathbf{S}_{bg}$
  Form intensity histograms from $\mathbf{S}_f$ and $\mathbf{S}_b$.
  Compute μ_i and σ_i, $i \in \{f, b\}$ from histograms.
  Compute decision rule per Eq. (1).
  if non-nadir phase **then**
    Foreground Mask: $\mathbf{S}_{mask} \leftarrow \mathcal{R}\left(\mathbf{S}_e\right) \bigcap \mathbf{S}_{fg}$
  else
    compute *fst+* per Algorithm 3
  if Resized **then**
    Resize foreground mask to the original input image size



(a) $|\mathcal{F}\left(\bar{\mathbf{I}}, \mathbf{S}_e\right) - \bar{S}_{fed}|$ (b) FED Result (c) $\overline{\mathcal{D}}\left(\mathbf{S}_e\right)$ (d) DoG Result

Fig. 2 FED vs. DoG Comparisons. Figure (a) and (c) are FED and DoG Response Maps, Figure (b) and (d) are FED and DoG Segmentation Results, Respectively.

separation in the image histogram as the threshold value. Figure 5 provides comparisons between thresholding and region detection, where Fig. 5(a) is the original image, subfigure (b) is the Otsu thresholding [51], subfigure (c) is

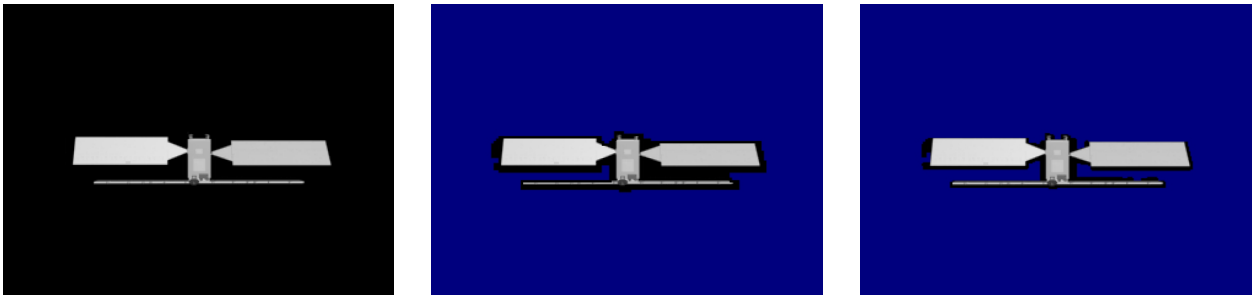


(a) $\bar{\mathbf{I}} \leftarrow \mathcal{N}(\mathbf{I})$ (b) $\mathbf{S}_M \leftarrow \mathcal{M}(\bar{\mathbf{I}})$ (c) $\mathbf{S}_u \leftarrow \mathcal{R}(\mathbf{S}_M)$ (d) $\mathbf{S}_e \leftarrow \nabla(\bar{\mathbf{I}})$ (e) $\mathbf{S}_d \leftarrow \overline{\mathcal{D}}(\mathbf{S}_e)$

(f) $\mathbf{S}_o \leftarrow \mathcal{R}(\mathcal{O}(\mathbf{S}_d))$ (g) $\mathbf{S}_m \leftarrow \mathcal{R}(\mathcal{M}(\mathbf{S}_d))$ (h) $\mathbf{S}_{fg} \leftarrow \mathbf{S}_u \bigcup \mathbf{S}_o \bigcup \mathbf{S}_m$ (i) $\mathbf{S}_{bg} \leftarrow \sim(\mathbf{S}_{fg})$ (j) $\mathbf{S}_f \leftarrow \bar{\mathbf{I}} \bigcap \mathbf{S}_{fg}$

(k) $\mathbf{S}_b \leftarrow \bar{\mathbf{I}} \bigcap \mathbf{S}_{bg}$ (l) FG/BG Histograms (m) $\mathbf{S}_{fe} \leftarrow \mathcal{R}(\mathbf{S}_e)$ (n) $\mathbf{S}_{fg} \leftarrow \mathbf{S}_{fe} \bigcap \mathbf{S}_{fg}$ (o) $\mathbf{I} \bigcap \mathbf{S}_{fg}$

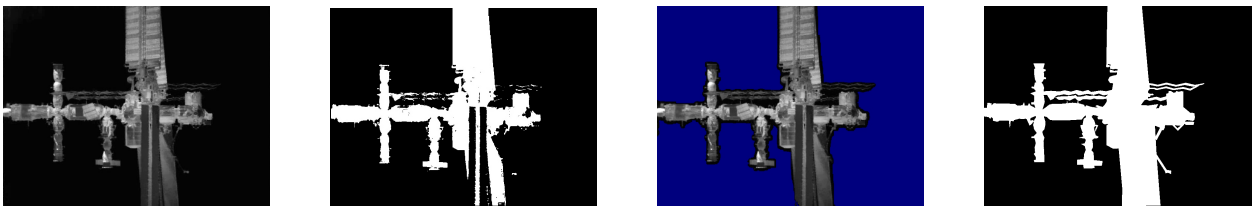
Fig. 3 Pointing Phase Identification and Foreground Mask Generation, Refer to Algorithm 1 for Details.



(a) Original (b) Without $\bigcap \mathbf{S}_{fe}$ (c) With $\bigcap \mathbf{S}_{fe}$

Fig. 4 Radarsat 3D Model Test Image. Original, Without and With Intersecting the Main Region to the Edge Response.

our region detection method in Algorithm 1 overlaid on top of the original image, the none blue region represents the foreground mask, and subfigure (d) is the ground truth (GT) mask. Our method slightly overpredicts the border region, but do not exclude any spacecraft regions.



(a) Original (b) Otsu (c) Ours (d) GT

Fig. 5 ISS IR Image Comparison of Contrast Thresholding and Region Detection. Comparing the Original Image, Otsu Response, Our Method and the Ground Truth.

B. Nadir Pointing

The nadir pointing Earth passage view is more difficult to process than the non-nadir pointing images. In addition to a cluttered background, part of the challenge comes from the single intensity channel of the IR image. In most cases, the foreground pixels are none distinctive from the background and therefore harder to classify. We develop two nadir pointing foreground extraction procedures with cluttered Earth background, these methods are purely image-driven and do not require pre-inference training. The first method uses Laplace operator response to extract the foreground, the second method is an accelerated manifold ranking technique.

1. Laplace Operator Response

We develop the *LAPLACE* algorithm as an efficient way to extract the high-frequency response by using the Laplacian and morphological operators. Let us define $\mathcal{CG}(\mathbf{x})$ as a function that converts a colour image into a standard greyscale image, and $\mathcal{B}(\mathbf{x})$ as the box filter operator. We also define $\mathcal{MD}(\mathbf{x})$ and $\mathcal{ME}(\mathbf{x})$ as the morphological dilation and erosion operators using an elliptical kernel respectively. The box filter and morphological operators are used to limit image noise and enlarge the high-frequency response regions. The *LAPLACE* algorithm is given in Algorithm 2, kernel sizes are selected based on sensitivity studies on dataset images from Sec. V.

Algorithm 2 The *LAPLACE* Foreground Mask Algorithm

procedure LAPLACE($\mathbf{I}$)
  $\mathbf{S}_L \leftarrow \mathcal{O}\left(\mathcal{N}\left(\left|\mathcal{B}\left(\mathcal{L}\left(\mathcal{CG}\left(\mathbf{I}\right), K_L = 3\right), K_B = 5\right)\right|\right)\right)$
  for $i = 1 : (K = 3)$ **do**
    $\mathbf{S}_L \leftarrow \mathcal{MD}\left(\mathbf{S}_L, K_d = 10\right)$
  for $i = 1 : (K = 2)$ **do**
    $\mathbf{S}_L \leftarrow \mathcal{ME}\left(\mathbf{S}_L, K_e = 10\right)$
  Foreground Mask: $\mathbf{S}_L \leftarrow \mathcal{R}\left(\mathbf{S}_L\right)$

2. Background on Graph Manifold Ranking

Before discussing our accelerated manifold ranking technique, we review background definitions of the Manifold Ranking or GMR method. The GMR method is a rating algorithm that spreads the seeding query scores to neighbouring nodes via a weighted network [52]. GMR has been widely adopted for document [53] and image retrievals [54]. The details of the original GMR algorithm, $\overline{GMR}(\mathbf{x})$, is provided by Yang *et al*. [41]. We will briefly describe key parameters and concepts of the baseline GMR algorithm to support discussions of our enhancements.

The input image is converted from *RGB* into the *CIE Lab* colour space using $\mathcal{CL}(\mathbf{x})$. We compute N number of Simple Linear Iterative Clustering (SLIC) [55] superpixels using the function $\mathcal{SL}(\mathbf{x})$. Each superpixel has a feature vector $\mathbf{h}^{(i)} \in \mathbb{R}^m$. We denote $\mathbf{f}$ as the ranking function for the superpixels. $\mathbf{z}$ is an indication vector, if $z^{(i)}$ is 1, we perform query on $\mathbf{h}^{(i)}$, else if $z^{(i)}$ is 0 then the query is not performed. We can use a *affinity matrix* $\mathbf{W} = [w^{(ij)}]$, $\mathbf{W} \in \mathbb{R}^{N \times N}$, and a *degree matrix* $\mathbf{D} = diag\{d^{(11)}, \ldots, d^{(NN)}\}$ to rank the queried superpixels. For any given

superpixel, its adjacent neighbours and the close-loop border boundaries are used for the feature-distance differencing. The GMR [41] ranking function, which represent optimal estimation of the foreground, is

$$\mathbf{f}^* = \mathbf{D}^{\frac{1}{2}}\tilde{\mathbf{A}}\mathbf{D}^{\frac{1}{2}}\mathbf{z} = \mathbf{D}^{\frac{1}{2}}\left(\mathbf{D} - \alpha\mathbf{W}\right)^{-1}\mathbf{D}^{\frac{1}{2}}\mathbf{z}, \tag{2}$$

were $\tilde{\mathbf{A}}$ is the unnormalised Optimal Affinity Matrix (OAM), α is manually tuned to 0.9. We take the normalised complementary vector for the foreground ranking score to be

$$\bar{\mathbf{f}}^* = \mathbf{1} - \frac{\mathbf{f}^*}{\underset{f^*}{\operatorname{argmax}}\, f^{*(i)}}, \tag{3}$$

where $f^{*(i)}$ is the ranking score at the i-th node. This ranking score is used to label individual superpixels in the final saliency map. The weighting is computed as $w^{(ij)} = \exp\left(\left(\check{D} - D^{(ij)}\right)/\delta\left(\hat{D} - \check{D}\right)\right)$, where $i,j \in \bar{\mathbf{V}}$ and $\bar{\mathbf{V}}$ contains all feature vectors from superpixels of the image, δ is manually tuned to 0.1. We use the L1-norm instead of the L2-norm for speed [56], $D^{(ij)} = \sum_{k\in m} |h_k^{(i)} - h_k^{(j)}|$, where the maximum and minimum feature vector difference is $\hat{D} = \arg\max_{i,j\in\bar{\mathbf{V}}} D^{(ij)}$ and $\check{D} = \arg\min_{i,j\in\bar{\mathbf{V}}} D^{(ij)}$ respectively. The background seeding taken from each of the four outer border superpixels is ranked by Eq. (2) and Eq. (3) to form the foreground saliency maps and are piece-wise multiplied. A mean value binary threshold is applied and the resulting normalised ranking scores are fed into Eq. (2) again. The ranking scores are transformed back to the saliency image space by $S_f^{(i)} = \bar{f}^{*(i)}$ where i is the node label index, $i \in \{1 \ldots N\}$.

3. Accelerated Manifold Ranking

Our accelerated manifold ranking increases the speed of GMR by more than 11 times and improves its precision performance. We use the best estimate foreground and background seeding rather than border seeding. The final GMR ranking computed by Eq. (2) is a function of the OAM, $\tilde{\mathbf{A}}$, and the estimated nodes of the foreground, $\mathbf{w}$, as queries. We replace the border background seeds with nodes under the best estimate background and foreground maps. Since the target object in spacecraft applications typically contains strong artificial edges compared to the softly blended background, the high-frequency edges are extracted by applying a 3×3 Laplacian filter, $\mathcal{L}(\mathbf{x})$, on the greyscale input image, $\mathbf{I}$, and blurred by a square box filter of size K_B to limit noise. We focus the foreground region from eye fixation attention cues computed by the Spectral Residual (SR) [57] method. The attention cue is combined with the high-frequency region by using a Gaussian distribution map, $\mathcal{G}(\mathbf{x})$, centred at the SR moment centroid, $\mathcal{MC}(\mathcal{SR}(\mathbf{I}))$. The *Hi*gh-frequency *Sa*liency *fe*ature (HiSafe), $\mathbf{S}_{hsf}$, is computed by normalising the combined SR response and the Laplace filtered response. We also leverage saliency map generated using a modified colour RC [39], $\mathcal{RC}(\mathbf{x})$, and MBD [40], $\mathcal{MB}(\mathbf{x})$. In addition, for a faster implementation of RC, we replace the region segmentation with the SLIC superpixels. Our experiments suggest the best estimate of the foreground map is the Otsu [51] threshold of

the Laplace response map, $\mathbf{S}_L$, unioned with the mean value binary threshold on the piece-wise multiplication of the Gaussian, HiSafe, MBD, and RC response maps as shown in step 14 of Algorithm 3, where grey coloured text is the *fst* algorithm such that $fst_flag ==$ TRUE. We compute both foreground and background GMR response and combine into one final saliency map, $\mathbf{S}_{fg}$, and apply a sigmoid function to enhance contrast [40]. The foreground mask can be computed by applying the mean binary threshold operation on $\mathbf{S}_{fg}$.

We designate the above procedure as the *fst* algorithm. It demonstrates a net increase in performance and significant speed improvements from the original GMR when evaluated on colour and greyscale images. It, however, still lacks the desired precision when applied to the IR images from the STS-135 ISS flight mission. An extended version of the *fst* algorithm is therefore developed to improve performance on spacecraft IR images. We designate this extended version as *fst+* in Algorithm 3, where the black coloured text is the extended *fst+* algorithm such that $fst_flag ==$ FALSE. The main addition to the extended *fst* model is by reusing the edge response maps computed from Algorithm 1. The DoG threshold responses add more confidence to the *fst* foreground prediction without requiring additional computational time. Finally, the best estimate foreground response is added to the GMR saliency response to adjust the coarse resolution output from the SLIC superpixels.

IV. Pose Estimation

The best estimate foreground mask described in the previous sections can be used as a prior, or to enhance a known prior input, or to simplify the input image for the region-based pose estimation. Our region-based pose estimation combines level-set segmentation and 3D model registration by minimising an energy function using 3D projection feedback and foreground-background pixel likelihood estimation. In this section, we describe how the previously computed foreground mask can be used with our modified level-set function for pose estimation purposes.

A. Notation Definitions and Rotation Transformations

Given an input image $\mathbf{I}$ and the image domain $\Omega \subset \mathbb{R}^2$. The image pixel $\mathbf{x}$ with coordinates (x, y) has a corresponding feature $\mathbf{y}$. This feature could be the pixel intensity or the colour vector (*e.g.* RGB, CIE Lab). Let us define C as the contour around the object of interest. The foreground region segmented by C is Ω_f, the background is Ω_b, an example is provided in Fig. 6(a), where the contour line of the Envisat image including definitions for its foreground and background regions. The foreground and background regions have their own statistical appearance model, $P(\mathbf{y}|M_i)$ for $i \in \{f, b\}$, where P is the probability density function, Φ is the level-set embedding function. More details of Φ shall be provided in Sec. IV.B. Finally, let $H(z)$ and $\delta(z)$ denote the smoothed Heaviside step function and the smoothed Dirac delta function respectively.

A 3D point $\mathbf{X}_c \in \mathbb{R}^3$ with coordinates $(X_c, Y_c, Z_c)^T$ expressed in the camera frame, $\overrightarrow{\mathcal{F}}_c$, can be a transformation of the object point $\mathbf{X}_b \in \mathbb{R}^3$ expressed in the object body frame, $\overrightarrow{\mathcal{F}}_b$, with coordinates $(X_b, Y_b, Z_b)^T$. Using a translation, $\mathbf{t}$, from $\overrightarrow{\mathcal{F}}_c$ to $\overrightarrow{\mathcal{F}}_b$ expressed in $\overrightarrow{\mathcal{F}}_c$ and a rotation, $\mathbf{R}$, from $\overrightarrow{\mathcal{F}}_b$ to $\overrightarrow{\mathcal{F}}_c$ that is parameterised by the quaternion

Algorithm 3 The *fst* and *fst+* Saliency Algorithm

procedure FST_FST_PLUS($\mathbf{I}$,fst_flag)
  $\mathbf{I}_{Lab} \leftarrow \mathcal{CL}(\mathbf{I})$
  $\mathbf{S}_{SR} \leftarrow \mathcal{N}(\mathcal{SR}(\mathbf{I}, \sigma = 2))$
  $\mathbf{S}_G \leftarrow \mathcal{G}(\mathcal{MC}(\mathbf{S}_{SR}))$
  $\mathbf{S}_L \leftarrow \mathcal{N}(|\mathcal{B}(|\mathcal{L}(\mathbf{I}, K_L = 3)|, K_B = 5)|)$
  $\mathbf{S}_{hsf} \leftarrow \mathcal{N}(\mathbf{S}_{SR} + \mathbf{S}_L)$
  $\mathbf{S}_{hsf} \leftarrow \mathbf{S}_{hsf} + Mean(\mathbf{S}_{hsf})$
  $\mathbf{S}_{mbd} \leftarrow \mathcal{N}(\mathcal{MB}(\mathbf{I}_{Lab}))$
  $\mathbf{S}_{rc} \leftarrow \mathcal{RC}(\mathbf{I}_{Lab})$
  if $fst_flag ==$ TRUE **then**
    $\mathbf{S}_{fg} \leftarrow \mathcal{O}(\mathbf{S}_L) \bigcup \mathcal{M}(\mathbf{S}_G \circ \mathbf{S}_{hsf} \circ \mathbf{S}_{mbd} \circ \mathbf{S}_{rc})$
  else
    $\mathbf{S}_{fg} \leftarrow \mathcal{O}(\mathbf{S}_L)/255$
    $\mathbf{S}_{fg} \leftarrow \mathbf{S}_{fg} + \mathcal{M}(\mathbf{S}_G \circ \mathbf{S}_{hsf} \circ \mathbf{S}_{mbd} \circ \mathbf{S}_{rc})/255$
    $\mathbf{S}_{fg} \leftarrow \mathbf{S}_{fg} + \mathbf{S}_o/255 + \mathbf{S}_m/255$ [a].
    $\mathbf{S}_p \leftarrow (\mathbf{S}_{fg} > 1)$.
  $\mathbf{S}_{fg} \leftarrow \mathcal{N}(\overline{\mathcal{GMR}}(\mathbf{S}_{fg}) \circ (255 - \overline{\mathcal{GMR}}(255 - \mathbf{S}_{fg})))$
  for $i = 1 : \mathbf{S}_{fg}$ height **do**
    for $j = 1 : \mathbf{S}_{fg}$ width **do**
      $\mathbf{S}_{fg}(i,j) \leftarrow \frac{1}{1+e^{-10(\mathbf{S}_{fg}(i,j)-0.5)}}$
  $\mathbf{S}_{fg} \leftarrow \mathcal{N}(\mathbf{S}_{fg})$
  if $fst_flag ==$ TRUE **then**
    Foreground Saliency Map: $\mathbf{S}_{fg}$
  else
    Foreground Saliency Map: $\mathbf{S}_{fg} \leftarrow \mathbf{S}_{fg} + \mathbf{S}_p$
  Foreground Mask: $\mathbf{S}_{mask} \leftarrow \mathcal{M}(\mathbf{S}_{fg})$

[a] where $\mathbf{S}_o$ and $\mathbf{S}_m$ are from Algorithm 1

$\mathbf{q} = (\boldsymbol{\epsilon}^T, \eta)^T = (q_x, q_y, q_z, q_w)^T$, such that

$$\mathbf{R} = \left(\eta^2 - \boldsymbol{\epsilon}^T\boldsymbol{\epsilon}\right)\mathbf{1} + 2\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T - 2\eta\boldsymbol{\epsilon}^\times, \tag{4}$$

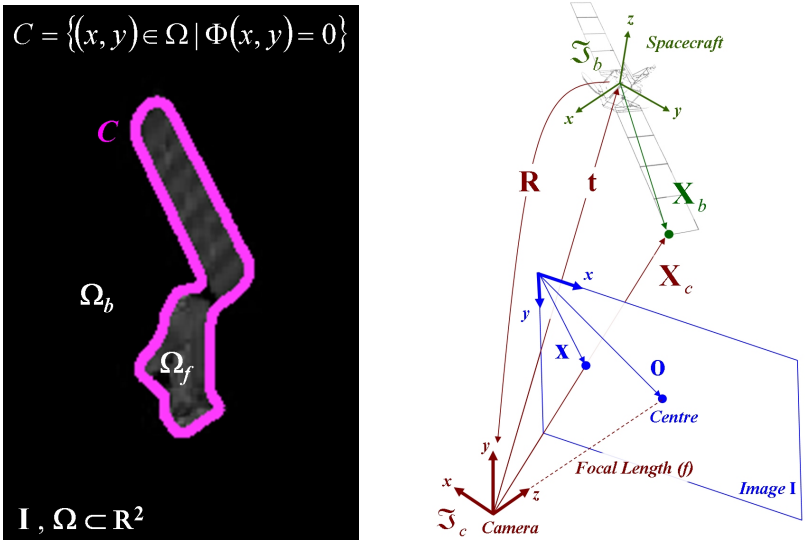
where $\mathbf{1} \in \mathbb{R}^{3\times3}$ is the identity matrix. Note the sign direction of q_w from Eq. (4) in our implementation. The individual coordinates of $(\mathbf{t}^T, \mathbf{q}^T)^T$ is represented by λ_i where $i \in \{1 \ldots 7\}$.

The camera is pre-calibrated by the intrinsic matrix

$$\mathbf{K} = \begin{bmatrix} fS_x & fS_\theta & o_x \\ 0 & fS_y & o_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{5}$$

where f is the focal length, S_θ is the pixel skew scaling, S_i and o_i are the image scale and centre coordinate for $i \in \{x, y\}$ respectively. The translation vector, $\mathbf{t}$, the rotation matrix, $\mathbf{R}$, the projection image, $\mathbf{I}$, the Spacecraft body

coordinate, $\mathcal{F}_b$, and the camera coordinate system, $\mathcal{F}_c$, are shown in Fig. 6(b).



(a) Region Definition (b) Coordinate Systems

Fig. 6 Spacecraft Image Contour, Level-set Function and Coordinate Systems

B. Level-set Pose Estimation

The level-set formulation [58] provides a simple mathematical framework for the implicit description of contour evolution. The merging, splitting, appearing, and disappearing of contours can be easily described by a higher dimensional entity Φ than by the explicit formulation of the curve entity C. The contour can be expressed explicitly as the 0th level in the level-set function Φ. For example, a contour in a 2D image is defined by the zero level in a Lipschitz continuous function Φ in a 3D surface. Formally, $C = \{(x,y) \in \Omega | \Phi(x,y) = 0\}$, the level-set function Φ is evolved rather than directly evolving C. An illustration of the level-set function evolution based on ISS motion is provided in Fig. 7, where the top row provides synthetic images of ISS motion with 3D model mesh projection overlay. The bottom row provides the corresponding level-set functions. The zero-level and zero crossing are indicated by gradient lines and magenta lines respectively. The zero-boundary is also projected above the zero-level as blue lines for clear illustration.

1. Segmentation Energy and Pixel Likelihood

The level-set formulation of the piecewise constant Mumford-Shah functional [59–61] that produces the two-phase segmentation of an image $I : \Omega \to \mathbb{R}$ by minimising an energy function [62]. Bibby and Reid [63] modified the energy formulation to use the likelihood of the pixel property $r_i(\mathbf{x}) = P(\mathbf{y}|M_i)$ for $i \in \{f,b\}$. The energy function is

$$\begin{aligned} E &= \int_{\Omega_f} r_f(\mathbf{I}(\mathbf{x}), C) d\Omega + \int_{\Omega_b} r_b(\mathbf{I}(\mathbf{x}), C) d\Omega \\ &= -\sum_{\mathbf{x} \in \Omega} \log\left(H(\Phi) P_f + (1 - H(\Phi)) P_b\right) \end{aligned} \tag{6}$$

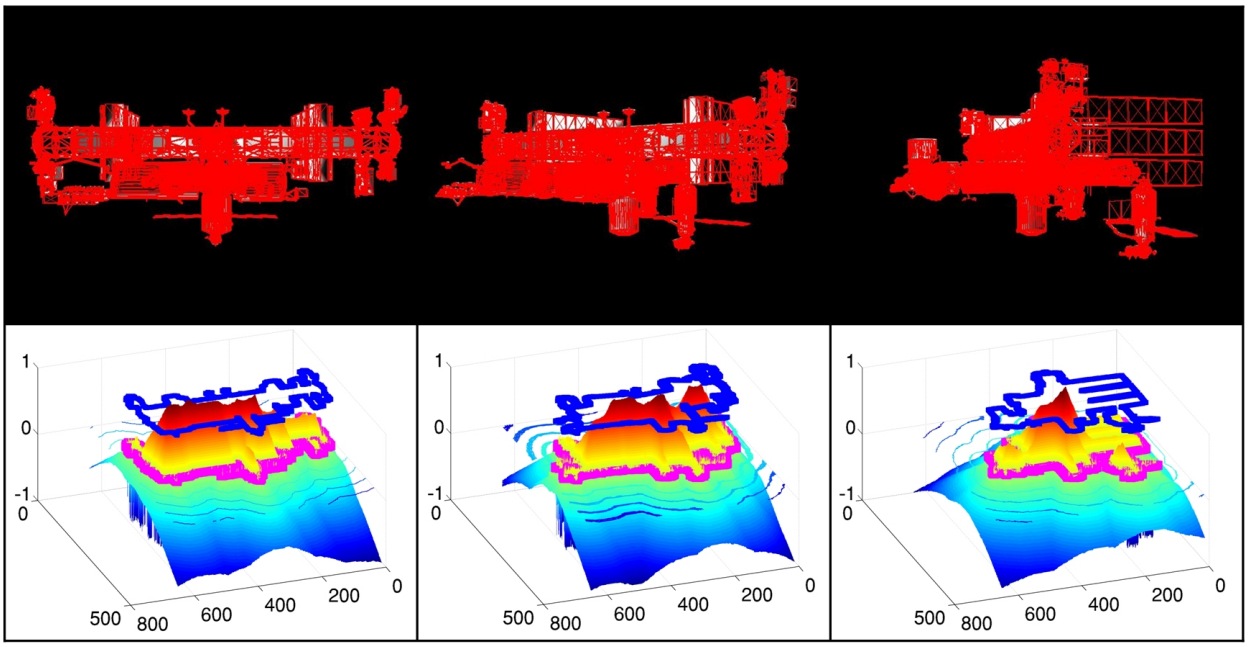


Fig. 7 ISS Pose Estimation Level-set Functions. Top: Image Registration Overlay Based on Computed Pose. Bottom: Level-set Function Zero Crossings in Magenta and Projected to Blue Lines for Visualisation.

where the respective probabilities are as follows, where $i \in \{f, b\}$.

$$P_i = \frac{P(\mathbf{y}|M_i)}{P(\mathbf{y}|M_f)\sum_{\mathbf{x}\in\Omega} H(\Phi(\mathbf{x})) + P(\mathbf{y}|M_b)\sum_{\mathbf{x}\in\Omega}(1 - H(\Phi(\mathbf{x})))}. \tag{7}$$

2. 3D Model Projection and Pose Estimation

The target object pose can be estimated using the energy functional as described in Eq. (6) by taking the partial derivative with respect to the individual pose parameters γ_i [46]; this allows the evolution of the target boundary with-respect-to its pose rather than time. Let us define $\partial(a)/\partial\gamma_i = a_{\gamma_i}$, $\nabla_t(a) = \left(a_{t_x}, a_{t_y}, a_{t_z}\right)^T$, and $\nabla_q(a) = \left(a_{q_x}, a_{q_y}, a_{q_z}, a_{q_w}\right)^T$. The energy partial-derivative is

$$E_{\gamma_i} = -\sum_{\mathbf{x}\in\Omega} \frac{P_f - P_b}{(P_f - P_b)H(\Phi) + P_b}\delta(\Phi)(\nabla\Phi)^T \mathbf{x}_{\gamma_i}, \tag{8}$$

where ∇ is the image gradient over $\mathbf{x}$. The camera projection model can be used to relate the 3D model to the 2D image as follows,

$$\begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} = Z_c^{-1}\mathbf{K}\begin{bmatrix} \mathbf{1} & \mathbf{0} \end{bmatrix}\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}\mathbf{X}_b = \mathbf{K}\frac{\mathbf{X}_c}{Z_c}, \tag{9}$$

where $\mathbf{0} \in \mathbb{R}^3$, $\mathbf{K}$ is the intrinsic camera matrix, $\mathbf{X}_c/Z_c$ is the depth normalised object point observed and expressed from the camera frame, f is the focal length of the camera, S_θ is the pixel skew scaling, S_i and o_i where $i \in \{x, y\}$ are

the pixel scaling and image origin to centre distance respectively. Equation (9) can be used to derive an expression for $\mathbf{x}_{\gamma_i}$ such that,

$$\mathbf{x}_{\gamma_i} = \frac{f}{Z_c^2} \begin{bmatrix} \mathbf{X}_c^T \left(S_x \mathbf{T}_x + S_\theta \mathbf{T}_y \right) \\ \\ \mathbf{X}_c^T \left(S_y \mathbf{T}_y \right) \end{bmatrix} (\mathbf{X}_c)_{\gamma_i}, \tag{10}$$

where $\mathbf{T}_i \in \mathbb{R}^{3\times 3}$, $i \in \{x, y\}$, has its elements equal to zero with the exception of $\mathbf{T}_x(3,1) = \mathbf{T}_y(3,2) = 1$ and $\mathbf{T}_x(1,3) = \mathbf{T}_y(2,3) = -1$. The partial derivative of $\mathbf{X}_c$ with-respect-to the pose parameters γ_i is derived from the extrinsic translation and rotation of the body coordinates to the camera coordinates through $\mathbf{X}_c = \mathbf{R}\mathbf{X}_b + \mathbf{t}$, the partial derivative results are as follows,

$$\nabla_t \mathbf{X}_c^T = \mathbf{1}, \quad \nabla_q \mathbf{X}_c^T = 2 \begin{bmatrix} \mathbf{A}\mathbf{X}_b & \mathbf{B}\mathbf{X}_b & \mathbf{C}\mathbf{X}_b \end{bmatrix}, \tag{11}$$

where $\mathbf{1}$ is the identity matrix and

$$\mathbf{A} = \begin{bmatrix} 0 & q_y & q_z \\ -2q_y & q_x & -q_w \\ -2q_z & q_w & q_x \\ 0 & q_z & -q_y \end{bmatrix}, \mathbf{B} = \begin{bmatrix} q_y & -2q_x & q_w \\ q_x & 0 & q_z \\ -q_w & -2q_x & q_y \\ -q_z & 0 & q_x \end{bmatrix}, \mathbf{C} = \begin{bmatrix} q_z & -q_w & -2q_x \\ q_w & q_z & -2q_y \\ q_x & q_y & 0 \\ q_y & -q_x & 0 \end{bmatrix}. \tag{12}$$

3. Centre Initialisation

Section IV.B.2 provides the connection between the model projection and level-set function using pixel probability. The gradient landscape surrounding the final pose minimum depends on the foreground and background pixel intensity variation. If the initial condition pose is specified in a region with black space background far from the target object, the gradient descent process can be highly sluggish. In this context, we develop a novel and simple initialisation scheme to avoid black space projection and reduce the number of steps that are required to reach the final pose potential minimum. Let us define the unaltered initial condition pose translation as $\tilde{\mathbf{t}}$, and our altered approach as $\mathbf{t}$. Figure 8 shows the original initial condition and our altered approach by using centralisation. The top figure in Fig. 8 shows the raw initial condition pose, captured image and actual target in grey. The bottom figure shows centralised initial condition pose using saliency mask geometric centroid (blue dot) to set the 3D model body frame projection. We use the computed saliency mask to generate a geometric centre of the region of interest. The pose translation image coordinate is set to the geometric centre, $(\bar{x}, \bar{y})$, which is computed using image areal moments. While there is no guarantee the body frame, which is normally located at the centre of mass, is the saliency mask geometric centre. It is likely this approximation will overlap the region of interest than an arbitrary chosen initial pose with a large

misalignment. This centre shift allows the gradient descent method to initiate in a region where the pose potential is more pronounced than if projecting the initial pose into a black space region with even gradients. The initial pose is computed as,

$$\mathbf{t} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{1}{fS_x}\left(Z_c\left(\bar{x} - o_x\right) - fS_\theta y\right) \\ \frac{Z_c}{fS_y}\left(\bar{y} - o_y\right) \\ Z_c \end{bmatrix}. \tag{13}$$

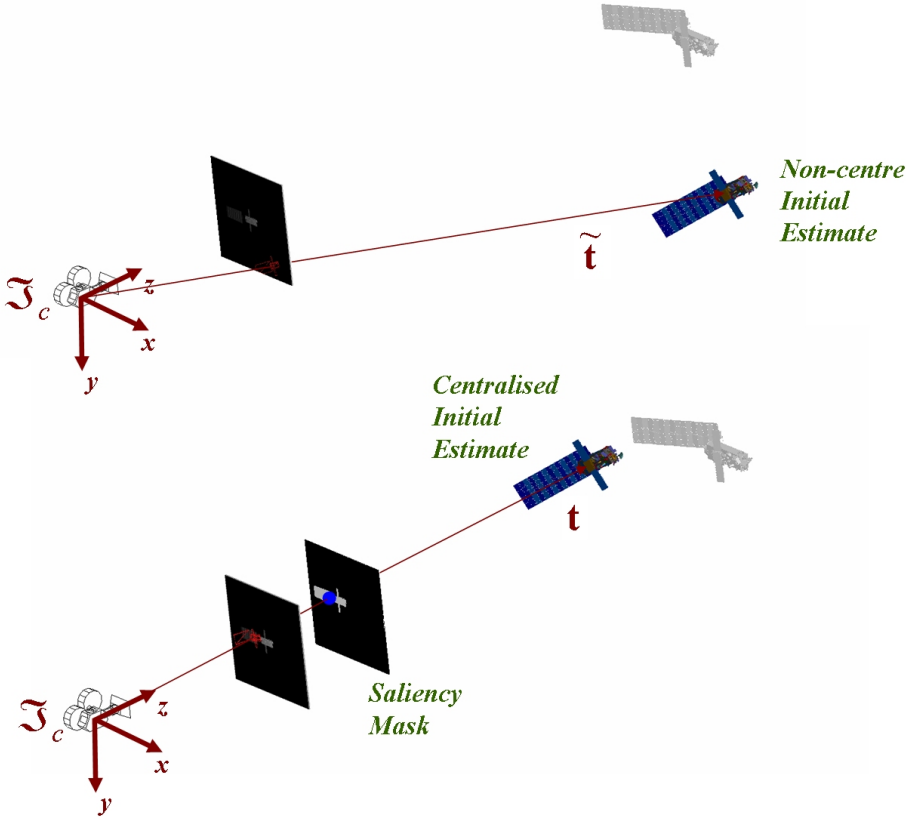


Fig. 8 Centre Initialisation of the Target Object. Top: Initial Non-centred Pose. Bottom: Applying Centred Formulation of Eq. (13) for Faster Gradient Descent.

4. Gradient Descent

The PWP3D [46] gradient descent method increments the pose parameter by manually adjusted stepsize. Let us define the pose vector as $\mathbf{x} = \begin{bmatrix} \mathbf{t}^T & \mathbf{q}^T \end{bmatrix}^T$. The baseline PWP3D gradient descent is,

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{h} \circ \mathbf{f}\left(\mathbf{x}_k\right), \tag{14}$$

where $\mathbf{h}$ is the step size for the individual pose axis, $\circ$ is the element-wise multiplication Hadamard product operator, $\mathbf{f}(\mathbf{x})$ is the gradient from Eq. (10) and q_{k+1} is L2-normalised after computing Eq. (14). The baseline gradient descent procedure is unstable and fails to produce the correct pose in several test cases. Specifically, we tested various gradient descent procedures including Nelder-Mead multi-dimensional simplex method [64], the combined Polack-Ribiere and Fletcher-Reeves method [46, 65]; however, neither method produced satisfactory timing and accuracy. Finally, we developed an enhanced gradient descent procedure with superior estimation results. The gradient magnitude variation directs the Nelder-Mead simplex [64] state change. The Polack-Ribiere and Fletcher-Reeves [46] step direction include the current and previous step gradients. Our tests show the current gradient direction outperforms both methods and is computationally more efficient and straightforward to implement. Unlike the baseline PWP3D, we use an alternative magnitude based on the inverse of the translation distance to modify the stepsize; because closer distance results in a larger image projection, and therefore require smaller gradient descent movement and vice-versa. Our improved gradient descent formulation is as follows,

$$\mathbf{t}_{k+1} = \mathbf{t}_k + \frac{h_t}{\|\mathbf{t}\|}\hat{\mathbf{f}}_t(\mathbf{x}_k), \tag{15}$$

where $\hat{\mathbf{f}}_t(\mathbf{x}_k)$ is the unit direction of the translational gradient, $\mathbf{f}_t/\|\mathbf{f}_t\|$, and h_t is the translational stepsize. The rotational gradient descent formulation is

$$\tilde{\mathbf{q}}_{k+1} = \mathbf{q}_k + \frac{h_q}{\|\mathbf{t}\|}\hat{\mathbf{f}}_q(\mathbf{x}_k), \tag{16}$$

where $\tilde{\mathbf{q}}$ is an non-normalised quaternion, and $\hat{\mathbf{f}}_t(\mathbf{x}_k)$ is the unit vector of the quaternion gradient. The final state vector is $\mathbf{x}_{k+1} = \begin{bmatrix} \mathbf{t}_{k+1}^T & \tilde{\mathbf{q}}_{k+1}^T/\|\tilde{\mathbf{q}}_{k+1}\| \end{bmatrix}^T$. Figure 9 provides the difference of using the basic gradient descent formulation and our enhanced method, where rows 1-2 provides Envisat pose overlay for frames 0, 150, and 300; rows 3-4 provides Radarsat pose overlay for frames 200, 250, and 300; rows 1 and 3 uses the PWP3D gradient descent implementation [46]; and rows 2 and 4 uses the enhanced gradient descent method. The original version destablised when the projection silhouette transitioned into the minimum region whereas the enhanced method remained stable through-out the entire estimation process.

V. Experimental Images and Datasets

In this section, we benchmark our saliency method against 14 traditional and recent saliency detection methods shown in Table 1. These methods were selected based on their real-time and near-real-time running time performance. There are many popular image datasets available for saliency and segmentation benchmark [39, 41, 74], however, there is no image dataset developed explicitly for spacecraft navigation applications. To facilitate our experiment and testing, we developed the Satellite Segmentation (SatSeg) dataset composed of 100 colour and greyscale spacecraft

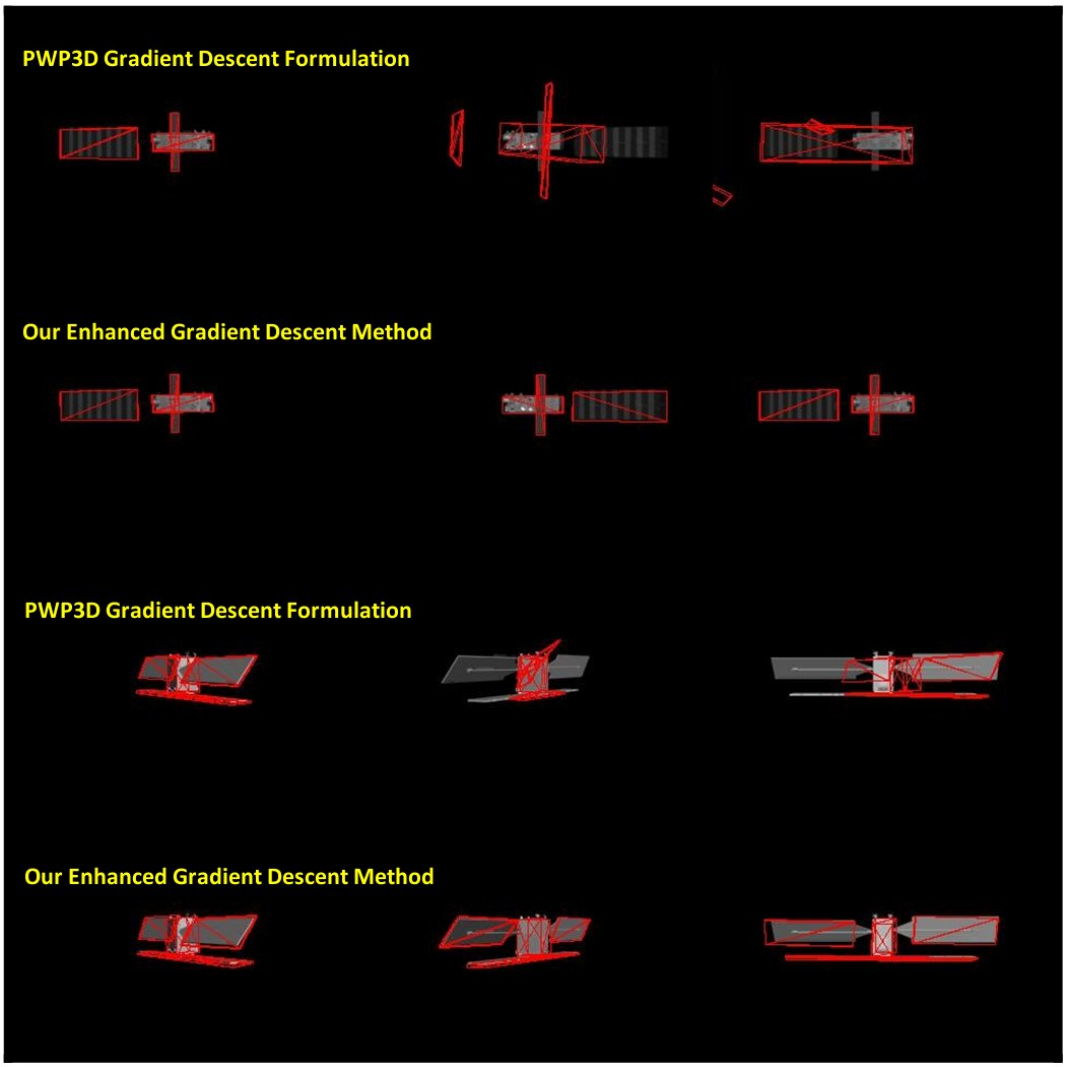


Fig. 9 Enhanced Gradient Descent Results for Envisat and Radarsat Synthetic Image Pose Estimation

images captured by photo and thermal cameras that are representative of the various mission scenarios ranging from flight to laboratory tests. Ground truth data were produced manually and is freely available to download from our project website[a].

We use the Receiver Operating Characteristics (ROC) [75] to evaluate our saliency method. For the computed saliency map, it is converted to an 8-bit binary mask image by applying a constant threshold. The precision and recall of a single image can be computed from the ground truth by performing piece-wise operation to the saliency maps [69]. A threshold ranging from 0 to 255 is used to control recall. We compute the standard units of measure for saliency evaluations including average and maximum F-Measure [70]. The Area Under the Curve (AUC) is from the ROC True Positive Rate versus False Positive Rate plot [69]. We used two computer platforms in the saliency evaluation, a Linux platform that is *AMD 8-Core-4.0GHz 32GB-RAM*, and a Windows platform that is *AMD 4-Core-1.5GHz 6GB-*

[a] http://ai-automata.ca/research/hisafe2.html

Table 1 Benchmark Methods

Code	Description	Reference
OTSU-75	OTSU Thresholding	[51]
LC-06	Colour Contrast	[39, 66]
SR-07	Spectral Residual	[57]
QFT-08	Phase Spectrum Quaternion Fourier Transform	[67]
AC-08	Local Contrast Raster Scan	[68, 69]
FT-09	Frequency Tuned	[70]
MSS-10	Maximum Symmetric Surround	[71]
HC-11	Histogram-based Contrast	[72]
GMR-13	Graph Manifold Ranking	[41]
GC-13	Global Cues	[73]
GD-15	Geodesic	[40]
MBD-15	Minimum Barrier Distance	[40]
MB+-15	MBD Extended	[40]
RC-15	Regional Contrast	[39]

RAM. Two platforms and operating systems were used as some of the benchmark methods are provided in Windows compiled executables.

We use a combination of synthetic CAD images and space flight IR images for evaluation. The CAD images include 3D models of the Envisat, the Radarsat, and the ISS model. The 3D models are also the pose estimation software internal models. We use 3D Studio Max®to generate synthetic videos from the 3D models which include some lighting and shadowing effects. We also use flight images from the STS-135 ISS undocking phases recorded by a SSO 813×604 MDA *TriDAR* thermal camera. For faster processing, the IR image resolution was reduced to 320×240 with estimated camera calibration properties of $fS_x = 752.517$, $fS_y = 752.517$, and $fS_\theta = 0$.

VI. Results and Discussions

The contribution of this work includes the creation of a novel saliency detection method to extract the target spacecraft from its background and using the spacecraft image to calculate its pose estimation. This section will discuss the saliency detection and pose estimation results separately.

A. Saliency Performance

Figure 10 provides qualitative comparisons of SatSeg saliency images generated by the *fst+* method, boxed in green, against traditional and state-of-the-art saliency models. In Figure 10 from left to right columns are: the input, Otsu thresholding (OTSU-75) [51], Spectral Residual (SR-07) [57], Graph Manifold Ranking (GMR-13) [41],

Minimum Barrier Distance (MB+-15) [40], and Regional Contrast (RC-15) [39], our *fst+* method from Algorithm 3 (in green box), and the Ground Truth (GT). The Otsu thrsholding [51] is not typically considered as a saliency detection model. However, it is nevertheless included in the comparison to demonstrate that although it can generate precise foreground map in the non-nadir pointing phase, it has the worst error in the nadir pointing phase when there is a cluttered background. The SR [57] model provides useful attention location cues but does not produce sufficient detail of the target object. GMR [41], RC [39], and MB+ [40] are state-of-the-art saliency methods that have good performance. However, both RC and GMR run-times are in the half-second range, and MB+ requires additional segmentation to precisely extract the foreground mask. Figure 10 shows our method having the least background error while maintaining the acceptable resolution of the foreground region.

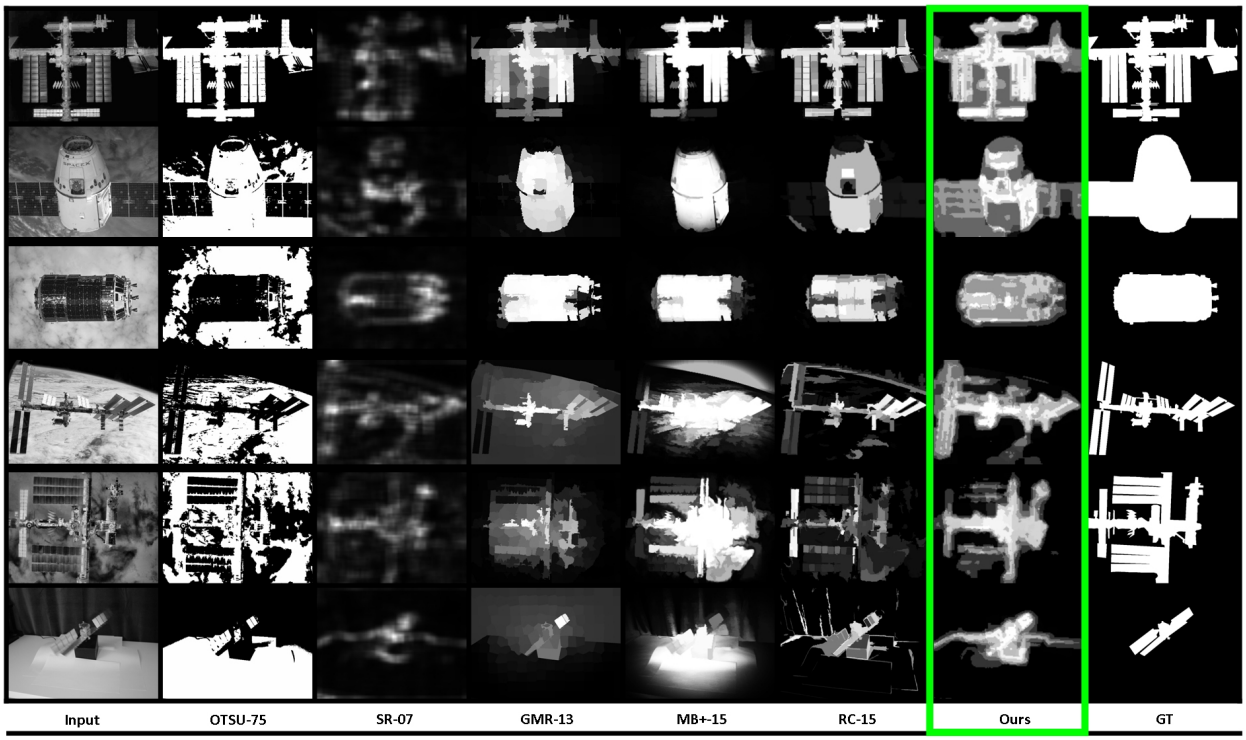
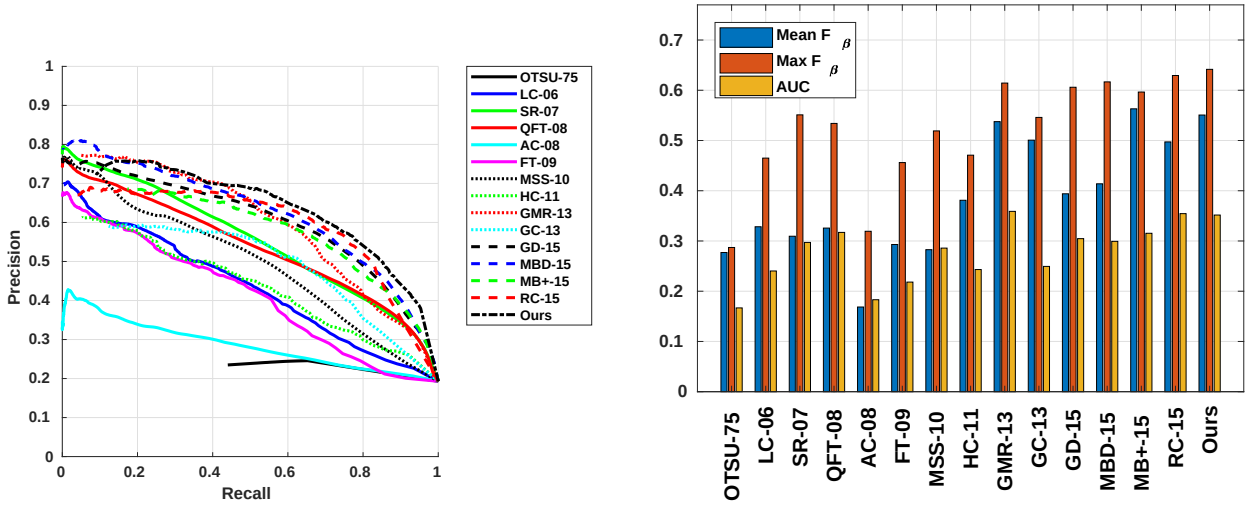


Fig. 10 Saliency Map Comparison of Selected Images from the Greyscale SatSeg Dataset

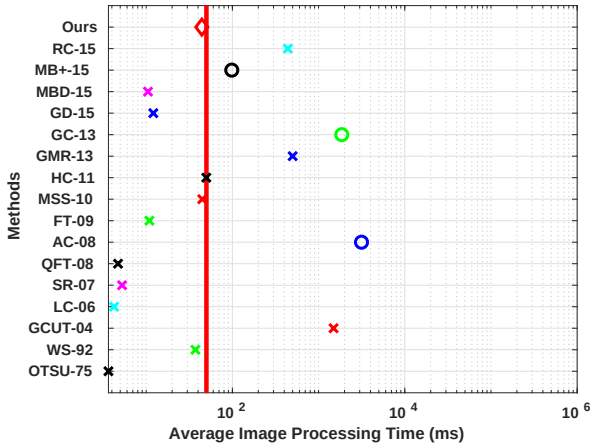
Quantitative performance plots are provided in Fig. 11, where subfigures (a) and (b) provides precision versus recall, mean F-measure, max F-measure, and AUC for 14 traditional and state-of-the-art saliency detection methods plus our *fst+* method respectively. Figure $11(c)$ provides the timing analysis for single image average run time of the SatSeg dataset; $\times$ and $\diamond$ marker runs are executed using C++/Linux platform; $\circ$ marker runs are executed using built executables on the Windows platform; the red line represents the design timing requirement for the saliency algorithm. Our *fst+* model shows the highest precision versus recall and maximum F-measure performance compared to all the methods. Our mean F-measure is second to RC and our AUC is comparable to RC and GMR. Speedwise, our *fst+* model average computation time for the SatSeg dataset is 44.223 ± 0.309 ms; this is $11\times$ faster than the original GMR, $10\times$ faster than RC, and $2.2\times$ faster than MB+. Faster methods such as MBD, GD, FT, QFT, SR, and LC has much

lower precision than our *fst+* method. We also compare with non-saliency detection methods such as Watershed (WS) [76], Graph Cut (GCUT) [77] and Otsu thrsholding (OTSU) [51]. In summary, Figs. 10 and 11 shows our *fst+* model to have the best performance overall.



(a) Precision vs. Recall

(b) f-Measure and AUC



(c) Timing

Fig. 11 Performance Comparison Plots for Image Saliency using the Greyscale SatSeg Dataset. Top: Saliency Precision Performance. Bottom: Saliency Speed Performance. Our method is *fst+* Algorithm 3.

In addition, we applied our saliency algorithms to the *TriDAR* thermal camera video from the STS-135 mission ISS undock and fly-by phase. Figure 12 shows the input video and the results of the various methods, where from left to right, the first column is the original ISS IR video; the second column is the foreground extraction using Algorithm 1 on the entire video; the third column is using Algorithm 1 during the non-nadir phase and using *fst* in Algorithm 3 during the nadir phase; the fourth column is using Algorithm 2 for the entire video; the fifth column is using Algorithm 1 during the non-nadir phase and using *fst+* in Algorithm 3 during the nadir phase. For the ISS IR video test, *fst+* in Algorithm 3 provides the best foreground extraction out of all techniques. The *REGION_DETECT* method over predicts while the *fst* method under predicts the foreground region. The *LAPLACE* method performed better than the more complex and relatively more expensive *REGION_DETECT* and *fst*. However, the *LAPLACE*

method can develop isolated holes or blocks as a result of the morphological operator kernel. The *fst+* method has the best foreground extraction performance overall; it can handle the majority of the IR video by accurately extracting only the ISS image. The only exception is in a section of the Earth passage (row 3 to 4 in Fig. 12) where the background developed sharp edges from the cloud regions; this section of the video is problematic for all the methods, where the ISS is mostly outside the viewable frame with only the solar panel being partially visible. The *fst+* method has the best prediction of the solar panels but over predicted the foreground region by including the cloud regions; this is due to a purely image-driven saliency detection, which is incapable of classifying useful and none useful high-frequency contents. Future work may include temporal data with some top-down guidance to increase precision.

The *fst+* effectiveness is demonstrated when it is combined with the region-based pose estimation. Figure 13 provides a single frame example of pose estimation during the nadir phase, where row 1 is the pose estimation using direct IR image input without any saliency detection. Row 2 is the normalised foreground and background histograms associated with the row 1 images. The first image on row 1 is the first input image with the projected initial misalignment. The first image on row 2 is the histogram template based on the prior mask. Row 3 is the pose estimation after applying the *fst+* saliency map with mean value binary threshold pre-processing. Row 4 is the foreground to background histograms associated with row 3. The gradient descent iteration steps associated with columns 2 to 5 are 0, 10, 30, and 60 respectively. The red lines in the histograms represent the foreground intensities, and the blue lines represent the background intensities. The initial translation and rotation misalignments are 17.3 meters, and 17.3 degrees RSS respectively. The saliency detection removes barriers from the gradient descent path to the true minimum solution. The background histogram in the second row has dimmer pixels than the background histogram in the fourth row; this is because both foreground and background histograms are normalised within its class. Indeed, the converged solution histogram for the fourth row is much closer to its template than the second row where the image without saliency detection is trapped in a local minimum.

B. Envisat Synthetic Image Pose Estimation

Next, we applied the region-based pose estimation technique on various spacecraft proximity operations scenarios including simulated Envisat tumbling and STS-135 ISS depart flight segments.

1. Envisat Synthetic Image Pose Estimation

The simulated Envisat motion includes rotation in the roll and yaw axis over two cycles. The Envisat is 80 meters away from the servicing spacecraft camera with a tumbling rate of 10 degrees per second. The camera resolution is 320×240 with calibrated intrinsic properties of $fS_x = 439.967$, $fS_y = 432.427$, and $fS_\theta = -0.0699286$. Figures 14 and 15 provides the pose estimation and estimation error of the resulting simulated Envisat tumbling motion. Figures 14(a) provides the greyscale image captured by a virtual camera in the simulation environment. The simplified 3D model is projected on to the image as visual overlays in red over one tumbling cycles. Subfigure (b) provides the

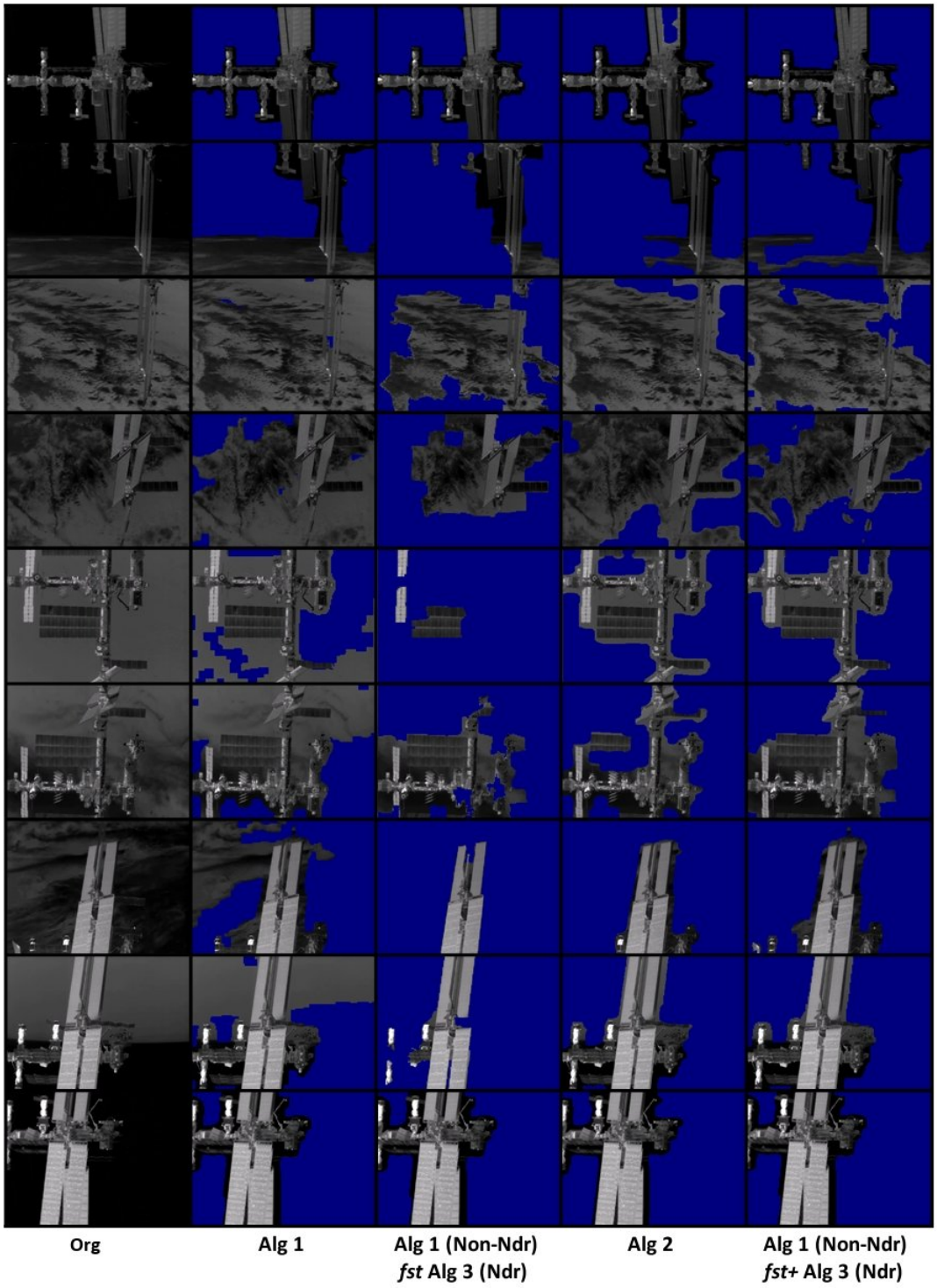


Fig. 12 ISS Saliency Method Comparisons

equivalent 3D plot of the Envisat motion computed using only the captured images compared to the ground truth (GT) in the Envisat spacecraft body frame. The position and orientation of the Envisat relative to the camera expressed in the camera frame is shown in Figs. 15(a) and (c), respectively, where red lines are the computed pose, blue lines are the ground truth. The position and orientation error is shown in Figs. 15 (b) and (d), respectively. The camera

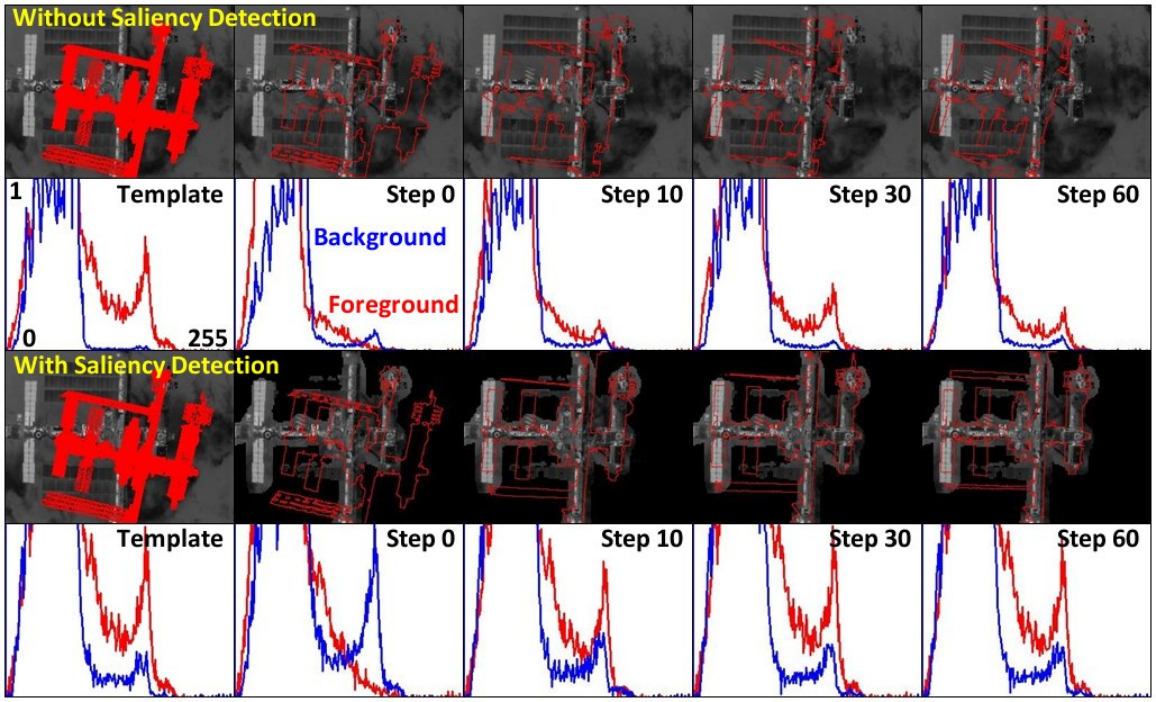
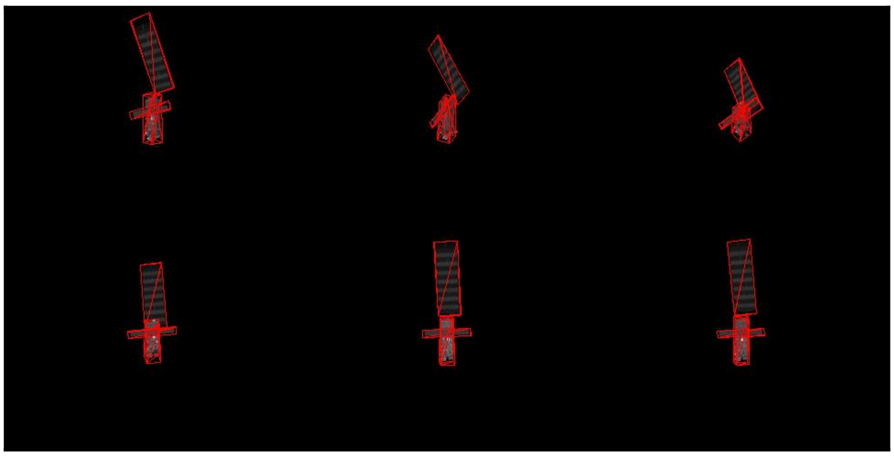
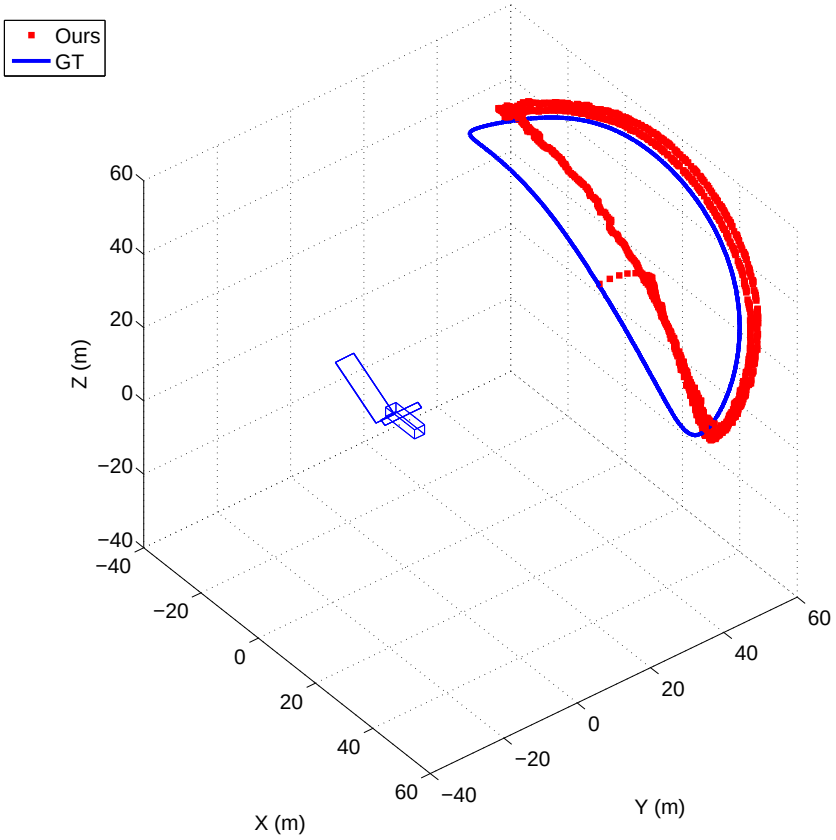


Fig. 13 STS-135 ISS Undocking and Flyby Pose Estimation During Earth Passage. Blue and Red Lines are Background and Foreground Normalised Histograms, Respectively.

pose is defined as the position from the camera frame to the Envisat body frame expressed in the camera frame, and orientation of the Envisat body frame rotated from the camera frame using the PYR Euler angle rotation sequence. The offset error in the X and Y direction is less than 0.1 and 0.5 meters, respectively. Out of 80 meters, total distance between the camera and Envisat, the Root-Sum-Squared (RSS) lateral error is 0.6 percent. There is a Z-axis drift of less than 4 meters or 5 percent. Figure 15(d) shows the orientation error is bounded within 15 and 6 degrees for roll and wobble, respectively. With respect to the absolute mean orientation, the approximate error is 10 and 3.3 percent for roll and wobble, respectively. It is not surprising the boresight error is larger than the lateral when using a single camera because without triangulation or time-of-flight measurement, the depth is dependent on the region area and image resolution, where a large distance between the target vehicle and the camera will result in a larger bore sight measurement error. An initial translational pose offset shown in Fig. 14(b) is caused by an orientation offset when expressed in the camera frame. Due to the RSS distance between the two vehicles being 80 meters, any small rotation error has been amplified into larger translation errors when expressed in the Envisat body frame. Figure 15(c) shows the orientation offset during peak oscillations. The orientation offset is largely due to the shadowing of the Envisat lower solar panel reducing the area of the spacecraft. This error highlights the sensitivity of the region-based method accuracy to the observable area.



(a) Pose Overlay

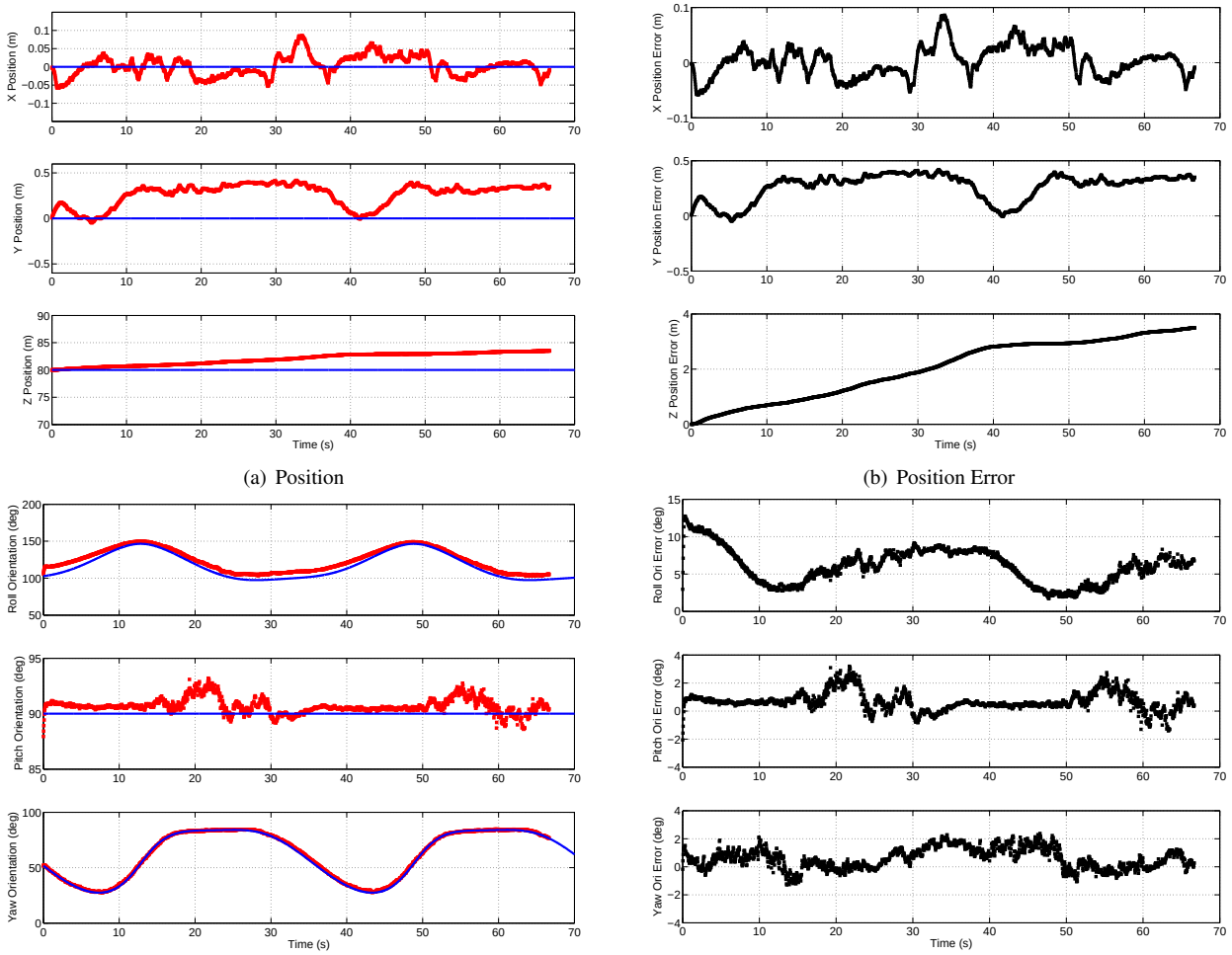


(b) 3D Pose

Fig. 14 Envisat Pose Estimation and Image Overlay. Top: Estimated Pose Image Registration Overlays. Bottom: Camera Pose Estimation 3D Plot Relative to the Target Envisat.

2. Pose Estimation Technique Comparison

We compare our region-based pose estimation method with a popular point-based method called *efficient Perspective-N-Point* (ePnP) [78]. The ePnP method computes the relative pose of an object by using geometric constraints from four virtual points generated using image features. For the pose estimation comparison, we use a combination of Scale Invariant Feature Transform (SIFT) [28] image feature points combined with homography transform, ePnP and



(a) Position (b) Position Error

(c) Orientation (d) Orientation Error

Fig. 15 Time Plot of Envisat Pose Estimation (Left) and Estimation Error (Right) Between Estimated Pose Computed from Monochromatic Monocular Camera Images and Ground Truth.

Random Sample and Consensus (RANSAC) [79] for the final pose estimation calculation. Details of the point-based estimation pipeline is provided in Shi *et al.* [80]. The main benefit of the region-based method is adding stability to the pose estimation compared to the point-based method. Some examples of the pose estimation are shown in Fig. 16[b]. The top figure shows the two methods computing the correct pose of the Envisat spacecraft, while in the bottom two images, the homography-ePnP method fails to produce the proper pose. Where the green lines and points are the initial step homography transform and the red lines and points are the ePnP matching. In the middle image, the homography transformed correctly and the ePnP failed to produce an acceptable pose, whereas, in the bottom image, both homography and ePnP failed to produce the right results. The main reasons are due to homography will not always be able to find a satisfactory transformation, and ePnP precision has dependencies on the amount of perspective observed from the image points. The proposed region-based method, on the other hand, is computed smoothly throughout the sequence because blob region is more stable than point features.

[b] https://youtu.be/8Km--FOmC8E

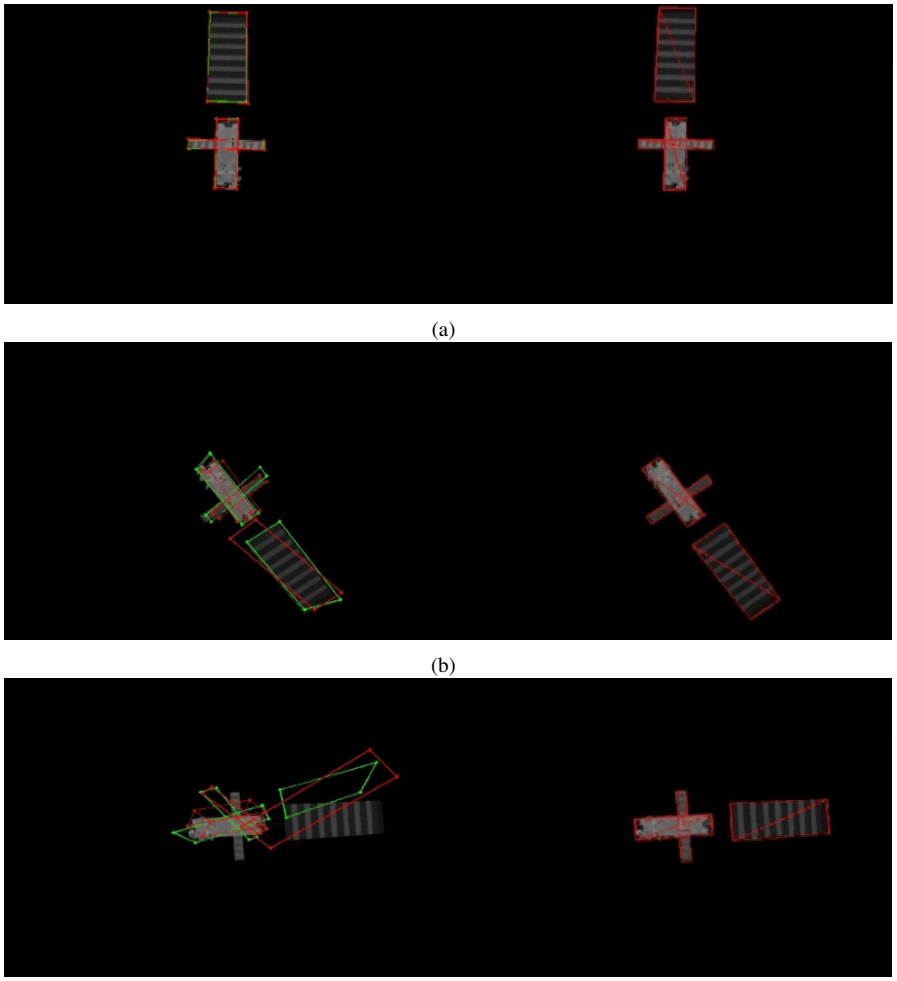


Fig. 16 Pose Estimation Comparison Between SIFT+ePnP (Left) and Our Proposed Method (Right). Feature Keypoint Efficient Prospective-N-Point Method is Less Stable than the Proposed Regional Method.

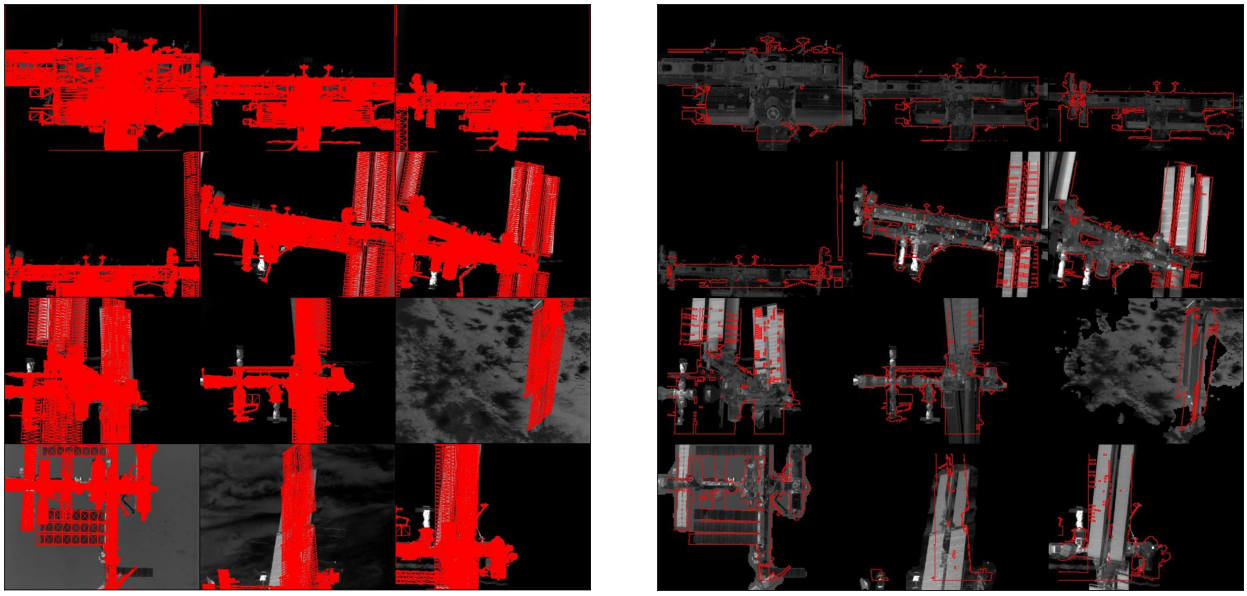
3. ISS Pose Estimation

The NASA ISS local orbital coordinate system or the Local-Vertical-Local-Horizontal (LVLH) [81] is in the general direction of ISS body frame with minor attitude offsets. Figures 17 shows the STS-135 ISS undocking and proximity flyby sequence IR image pose estimation; it provides the six degrees-of-freedom motion sequence of the SSO undocking from the ISS initially departing along the *foward* V-bar (LVLH X), then turns to the negative H-bar (LVLH Y) and performing a proximity flyby *overhead* the negative R-bar (LVLH $-Z$) from ISS *starboard* to *port* [82]. The *fst+* saliency mask is applied to the input IR image. Figure 18 provides the 3D reconstruction of the camera trajectory expressed in the ISS Body Coordinate System (ISSBCS) [81].

The initial condition estimated pose need to be reinitialised every ten frames on average to avoid error build-up. Certain phases of the proximity operations are more robust than others; for example, the initial V-bar departure, the negative R-bar overhead transition can be estimated more precisely than the V-bar to H-bar turn, the turn to R-bar, and turning from R-bar to H-bar. The V-bar departure and R-bar transition are mostly translational motion while

the turning sequences require changing the roll and pitch-axis Euler angles as shown in Fig. 18, where the camera frame with respect to the ISSBCS is shown in red squares; a to-scale simplified ISS model outline is shown in blue lines; the ISSBCS is located at the ISS mass centre and is in the general direction of the ISS LVLH coordinate system with slight attitude offsets. The pose estimation error is mainly caused by unknown ISS subcomponent configuration; for example, the solar panel pan and tilt angles, the SSRMS joint configuration, the radiation panel orientations, the Soyuz module attachment location, shape, and their solar panel deployment positions. All these unknowns add to the projection error and reduce the odds of achieving a perfect alignment. Secondly, in some frames, the ISS is not in full view. The regional method performs best when the entire vehicle is displayed in the image with distinctive shape features. Conversely, when the projected region almost entirely covers the image such as frames 1 and 366 in Fig. 17 or only a tiny portion of the vehicle is displayed such as frames 585 and 731, the regional method performs poorly due to ambiguity in the region silhouette. In Fig. 17 sequence frames 1, 74, 147, 220, 293, 366, 439, 512, 585, 658, 731, and 793 are displayed respectively. Subfigure (a) is the projection overlay of the 3D model on the ISS IR image. Subfigure (b) is the silhouette outline of the projection overlay while using the *fst+* saliency detection.

Based on the experiment results, it is recommended to include additional correction methods to increase prediction accuracy such as using feature localisation when the target is near, and there are an abundant amount of image features within the boundary region that can be used to clarify silhouette ambiguity. Secondly, the pose estimation precision can improve by combining stochastic filters with the camera pose estimation, taking into account predictions in the dynamic motion. This can be done, namely, by implementing a Kalman filter. Local histogram template matching may possibly also reduce region ambiguity as demonstrated by Hexner *et al*. [48] and Tjaden *et al*. [50].



(a) Pose Overlay (b) Projected Silhouette

Fig. 17 STS-135 ISS Undocking and Proximity Flyby Sequence Pose Estimation. Left: Estimated Pose 3D Projected Image Overlay. Right: Level-Set Function Silhouette Overlay.

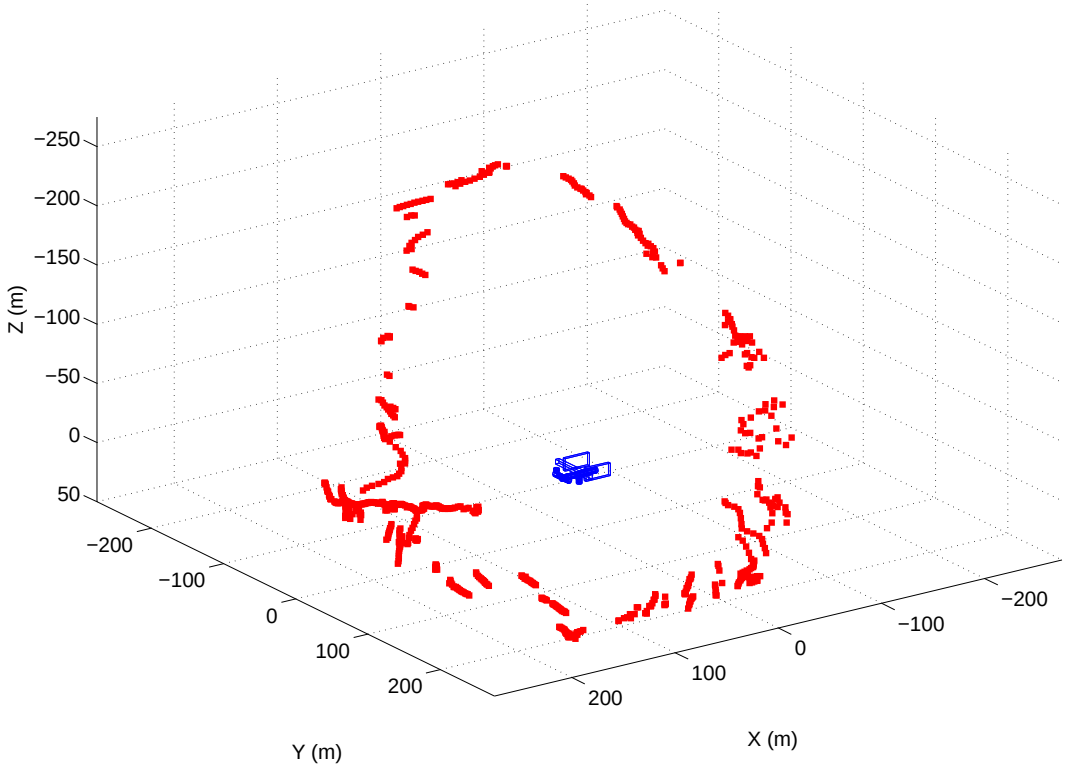


Fig. 18 STS-135 ISS Undocking and Proximity Flyby 3D Reconstruction From Monochromatic Monocular Camera Only Pose Estimation.

VII. Conclusion

In this paper, we provided a novel image-driven approach to distinguish nadir and non-nadir camera pointing. Specifically, we developed several saliency detection models including the *LAPLACE* foreground mask algorithm, the *fst* and *fst+* enhanced graph manifold ranking model for working with infrared greyscale images. Our evaluation of the *fst+* model shows performance exceeding traditional and state-of-the-art saliency detection methods, and our method is $11\times$ faster than the original graph manifold ranking method. Using these new algorithms, we then presented an end-to-end real-time spacecraft state generation model that includes saliency foreground extraction and region-based pose estimation using level-set segmentation and pixel statistics. We introduced innovative initialisation and gradient descent formulations for stable and efficient pose estimation convergence. Finally, we successfully demonstrated using this approach for pose estimation of a complex six degrees-of-freedom uncooperative target spacecraft motion. Future work include the improvement of the robustness of this pose estimation method by using stochastic filters and dynamic model predictions.

Acknowledgments

This research was jointly funded by the Natural Sciences and Engineering Research Council Scholarship CGSD3-453738-2014, Canadian Space Agency Space Technology Development Program and Ontario Centres of Excellence

Voucher for Innovation and Productivity II Award 24053.

References

[1] Ruel, S., Luu, T., and Berube, A., "Space Shuttle Testing of the TriDAR 3D Rendezvous and Docking Sensor," *Journal of Field Robotics*, Vol. 29, No. 4, 2012, pp. 535–553.

[2] Liu, C. and Hu, W., "Relative Pose Estimation for Cylinder-shaped Spacecrafts using Single Image," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 50, No. 4, 2014, pp. 3036–3056.

[3] Ventura, J., Fleischner, A., and Walter, U., "Pose Tracking of a Noncooperative Spacecraft During Docking Maneuvers using a Time-of-Flight Sensor," *AIAA Guidance, Navigation and Control Conference and Exhibit*, AIAA, San Diego, CA, Jan 2016.

[4] Zhang, L., Zhu, F., Hao, Y., and Pan, W., "Optimization-based Non-cooperative Spacecraft Pose Estimation using Stereo Cameras During Proximity Operations," *Applied Optics*, Vol. 56, No. 15, May 2017, pp. 4522–4531.

[5] Casonato, G. and Palmerini, G., "Visual Techniques Applied to the ATV ISS Rendezvous Monitoring," *IEEE Aerospace Conference Proceedings*, Vol. 1, Mar 2004.

[6] Mokuno, M. and Kawano, I., "In-orbit Demonstration of an Optical Navigation System for Autonomous Rendezvous Docking," *AIAA Journal of Spacecraft and Rockets*, Vol. 48, No. 6, Nov–Dec 2011, pp. 1046–1054.

[7] Wu, Y., Yang, G., Lin, J., Raus, R., Zhang, S., and Watt, M., "Low-Cost, High-Performance Monocular Vision System for Air Bearing Table Attitude Determination," *AIAA Journal of Spacecraft and Rockets*, Vol. 51, No. 1, Jan-Feb 2014, pp. 66–75.

[8] Zhang, G., Kontitsis, M., Filipe, N., Tsiotras, P., and Vela, P., "Cooperative Relative Navigation for Space Rendezvous and Proximity Operations using Controlled Active Vision," *Journal of Field Robotics*, Vol. 00, No. 0, 2015, pp. 1–24.

[9] Fourie, D., Tweddle, B., Ulrich, S., and Saenz-Otero, A., "Flight Results of Vision-Based Navigation for Autonomous Spacecraft Inspection of Unknown Objects," *AIAA Journal of Spacecraft and Rockets*, Vol. 51, No. 6, Nov-Dec 2014, pp. 2016–2026.

[10] Lim, T., P.F., R., and M.C., O., "Edge Detection using Point Cloud Data for Noncooperative Pose Estimation," *AIAA Journal of Spacecraft and Rockets*, Vol. 54, No. 2, Mar-Apr 2016, pp. 500–505.

[11] Lourakis, M. and Zabulis, X., "Satellite Visual Tracking for Proximity Operations in Space," *Proceedings of 14th ESA Workshop on Advanced Space Technologies for Robotics and Automation*, Leiden, the Netherlands, Jun 2017.

[12] Sharma, S., Ventura, J., and S., D., "Robust Model-Based Monocular Pose Initialization for Noncooperative Spacecraft Rendezvous," *AIAA Journal of Spacecraft and Rockets*, Vol. 55, No. 6, Nov-Dec 2018, pp. 1414–1429.

[13] Curtis, D. and Cobb, R., "Satellite Articulation Tracking Using Computer Vision," *AIAA Journal of Spacecraft and Rockets*, Vol. 56, No. 5, Sep-Oct 2019, pp. 1478–1491.

[14] Bodin, P., Larsson, R., Nilsson, F., Chasset, C., Noteborn, R., and Nylund, M., "PRISMA: An In-Orbit Test Bed for Guidance, Navigation, and Control Experiments," *AIAA Journal of Spacecraft and Rockets*, Vol. 46, No. 3, May-Jun 2009, pp. 615–623.

[15] Modenini, D., "Five-Degree-of-Freedom Pose Estimation from an Imaged Ellipsoid of Revolution," *AIAA Journal of Spacecraft and Rockets*, Vol. 56, No. 3, May-Jun 2019, pp. 952–958.

[16] Christian, J., "Accurate Planetary Limb Localization for Image-based Spacecraft Navigation," *AIAA Journal of Spacecraft and Rockets*, Vol. 54, No. 3, May-Jun 2017, pp. 708–730.

[17] Shi, J., Ulrich, S., and Ruel, S., "Unsupervised Method of Infrared Spacecraft Image Foreground Extraction," *AIAA Journal of Spacecraft and Rockets*, Vol. 56, No. 6, Nov-Dec 2019, pp. 1847–1856.

[18] Liu, Z., Xiang, Q., Tang, J., Wang, Y., and Zhao, P., "Robust Salient Object Detection for RGB Images," *The Visual Computer Journal*, Dec 2019.

[19] Huang, K. and Gao, S., "Image Saliency Detection via Multi-scale Iterative CNN," *The Visual Computer Journal*, 2019.

[20] Kisantal, M., Sharma, S., Park, T., Izzo, D., M., M., and S., D., "Satellite Pose Estimation Challenge: Dataset, Competition Design and Results," *IEEE Transactions on Aerospace and Electronic Systems*, Apr 2020.

[21] Rosenhahn, B., Perwass, C., and Sommer, G., "Foundations About 2D-3D Pose Estimation," *CVonline: The Evolving Distributed, Non-Proprietary, On-line Compendium of Computer Vision*, Edinburgh, 2004.

[22] Lepetit, V. and Fua, P., "Monocular Model-based 3D Tracking of Rigid Objects: A Survey," *Foundations and Trends in Computer Graphics and Vision*, Vol. 1, No. 1, 2005, pp. 1–89.

[23] Capuano, V., Ryan Alimo, S., Ho, A., and Chung, S., "Robust Features Extraction for On-board Monocular-based Spacecraft Pose Acquisition," *Proceedings of the AIAA Guidance, Navigation, and Controls Conf. and Exhibit*, San Diego, CA, Jan 2019.

[24] He, K., Zhang, X., Ren, S., and Sun, J., "Deep residual learning for image recognition," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015.

[25] Szegedy, C., Ioffe, S., and Vanhoucke, V., "Inception-v4, Inception-ResNet and the Impact of Residual Connections on Learning," *arXiv preprint*, arXiv:1602.07261, 2016.

[26] Harris, C. and Stephens, M., "A Combined Corner and Edge Detector," *Proceedings of the 4th Alvey Vision Conference*, Vol. 15, 1988, pp. 147–151.

[27] Rosten, E. and Drummond, T., "Machine Learning for High-speed Corner Detection," *European Conference on Computer Vision*, 2006, pp. 430–443.

[28] Lowe, D., "Distinctive Image Features from Scale-invariant Keypoints," *International Journal of Computer Vision*, Vol. 60, No. 2, 2004, pp. 91–110.

[29] Alcantarilla, P., Nuevo, J., and Bartoli, A., "Fast Explicit Diffusion for Accelerated Features in Nonlinear Scale Spaces," *British Machine Vision Conference*, Bristol, UK, Sept 2013.

[30] Choi, C. and Christensen, H., "RGB-D Object Pose Estimation in Unstructured Environments," *Robotics and Autonomous System*, Vol. 75, 2016, pp. 595–613.

[31] Zivkovic, Z. and van der Heijden, F., "Efficient Adaptive Density Estimation per Image Pixel for the Task of Background Subtraction," *Pattern Recognition Letters*, Vol. 27, No. 7, 2006, pp. 773–780.

[32] Godbehere, A., Matsukawa, A., and Goldberg, K., "Visual Tracking of Human Visitors Under Variable-lighting Conditions for a Responsive Audio Art Installation," *American Control Conference*, 2012, pp. 4305–4312.

[33] Long, J., Shelhamer, E., and Darrel, T., "Fully Convolutional Networks for Semantic Segmentation," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 3431–3440.

[34] Badrinarayanan, V., Handa, A., and Cipolla, R., "Segnet: A Deep Convolutional Encoder-decoder Architecture for Robust Semantic Pixel-wise Labelling," *Computing Research Repository*, Vol. abs/1505.07293, 2015.

[35] Yang, J. and Yang, M., "Top-down Visual Saliency via Joint CRF and Dictionary Learning," *IEEE Transactions on Pattern Analysis Machine Intelligence*, Vol. 39, No. 3, 2017, pp. 576–588.

[36] Li, L., Zhou, F., Zheng, Y., and Bai, X., "Saliency Detection Based on Foreground Appearance and Background-prior," *Neurocomputing*, Vol. 301, 2018, pp. 46–61.

[37] Krizhevsky, A., Sutskever, I., and Hinton, G., "Imagenet Classification with Deep Convolutional Neural Networks," *Neural Information Processing Systems*, 2012.

[38] Li, J., Tian, Y., Chen, X., and Huang, T., "Measuring Visual Surprise Jointly from Intrinsic and Extrinsic Contexts for Image Saliency Estimation," *International Journal of Computer Vision*, Vol. 120, No. 1, 2016, pp. 44–60.

[39] Cheng, M., Mitra, N., Huang, X., Torr, P., and Hu, S., "Global Contrast Based Salient Region Detection," *IEEE Transactions on Pattern Analysis Machine Intelligence*, Vol. 37, No. 3, 2015, pp. 569–582.

[40] Zhang, J., Sclaroff, S., Lin, Z., Shen, X., Price, B., and Měch, "Minimum Barrier Salient Object Detection at 80 FPS," *IEEE International Conference on Computer Vision*, 2015.

[41] Yang, C., Zhang, L., Lu, H., Ruan, X., and Yang, M., "Saliency Detection via Graph-based Manifold Ranking," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2013, pp. 3166–3173.

[42] Strand, R., Ciesielski, K., Malmberg, F., and Saha, P., "The Minimum Barrier Distance," *Computer Vision and Image Understanding*, Vol. 117, No. 4, 2013, pp. 429–437.

[43] Jung, C., Kim, W., Yoo, S., and Kim, C., "A Novel Monochromatic Cue for Detecting Regions of Visual Interest," *Journal of Image and Vision Computing*, Vol. 32, 2014, pp. 405–413.

[44] Yacoob, Y. and Davis, L., "Segmentation using Meta-texture Saliency," *IEEE International Conference on Computer Vision*, 2007.

[45] Dambreville, S., Sandhu, R., Yezzi, A., and Tannenbaum, A., "A Geometric Approach to Joint 2D Region-based Segmentation and 3D Pose Estimation using a 3D Shape Prior," *Journal on Imaging Sciences, Society for Industrial and Applied Mathematics*, Vol. 3, No. 1, 2010, pp. 110–132.

[46] Prisacariu, V. and Reid, I., "PWP3D: Real-time Segmentation and Tracking of 3D Objects," *International Journal of Computer Vision*, Vol. 98, No. 3, 2012, pp. 335–354.

[47] Perez-Yus, A., Puig, L., Lopez-Nicolas, G., Guerrero, J., and Fox, D., "RGB-D Based Tracking of Complex Objects," *International Workshop on Understanding Human Activities Through 3D Sensors*, Cancún, México, Dec 2015.

[48] Hexner, J. and Hagege, R., "2D-3D Pose Estimation of Heterogeneous Objects using a Region Based Approach," *International Journal of Computer Vision*, Vol. 118, No. 1, 2016, pp. 95–112.

[49] Tjaden, H., Schwanecke, U., and Schömer, E., "Real-time Monocular Segmentation and Pose Tracking of Multiple Objects," *European Conference on Computer Vision*, 2016, pp. 423–438.

[50] Tjaden, H., Schwanecke, U., and Schömer, E., "Real-Time Monocular Pose Estimation of 3D Objects using Temporally Consistent Local Color Histograms," *IEEE International Conference on Computer Vision*, 2017, pp. 124–132.

[51] Otsu, N., "A Threshold Selection Method from Gray-level Histograms," *Automatica*, 1975.

[52] Zhou, D., Weston, J., and Gretton, A., "Ranking on Data Manifolds," *Neural Information Processing Systems*, 2004.

[53] Wan, X., Yang, J., and Xiao, J., "Manifold-ranking Based Topic-focused Multi-document Summarization," *Proceedings of International Joint Conference on Artificial Intelligence*, 2007, pp. 2903–2908.

[54] Xu, B., Bu, J., Chen, C., Cai, D., He, X., Liu, W., and Luo, J., "Efficient Manifold Ranking for Image Retrieval," *ACM Special Interest Group on Information Retrieval*, 2011.

[55] Achanta, R., Shaji, A., Smith, K., Lucchi, A., Fua, P., and Süsstrunk, "SLIC Superpixels Compared to State-of-the-Art Superpixel Methods," *IEEE Transactions on Pattern Analysis Machine Intelligence*, Vol. 34, No. 11, 2012, pp. 2274–2281.

[56] Margolin, R., Tal, A., and Zelnik-Manor, L., "What Makes a Patch Distinct?" *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2013, pp. 1139–1146.

[57] Hou, X. and Zhang, L., "Saliency Detection: A Spectral Residual Approach," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2007.

[58] Osher, S. and Sethian, J., "Fronts Propagating with Curvature-Dependent Speed: Algorithms Based on Hamilton-Jacobi Formulations," *Journal of Computational Physics*, Vol. 79, No. 1, 1988, pp. 12–49.

[59] Mumford, D. and Shah, J., "Optimal Approximations by Piecewise Smooth Functions and Associated Variational Problems," *Communications Pure Applied Math*, Vol. 42, 1989, pp. 577–685.

[60] Tsai, A., A., Y., Wells, W., Tempany, C., D., T., Fan, A., Grimson, E., and Willsky, A., "Model-Based Curve Evolution Technique for Image Segmentation," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Kauai, Hawaii, 2001, pp. 463–468.

[61] Chan, T. and Vese, L., "Active Contours Without Edges," *IEEE Transactions on Image Processing*, 2001, pp. 266–277.

[62] Cremers, D., Rousson, M., and Deriche, R., "A Review of Statistical Approaches to Level Set Segmentation: Integrating Color, Texture, Motion and Shape," *International Journal of Computer Vision*, Vol. 72, No. 2, 2007, pp. 195–215.

[63] Bibby, C. and I., R., "Robust Real-Time Visual Tracking using Pixel-wise Posteriors," *European Conference on Computer Vision*, Berlin, Heidelberg, 2008, pp. 831–844.

[64] Nelder, J. and Mead, R., "A Simplex Method for Function Minimization," *The Computer Journal*, Vol. 7, No. 4, 1965, pp. 308–313.

[65] Hager, W. and Zhang, H., "A New Conjugate Gradient Method with Guaranteed Descent and an Efficient Line Search," *SIAM Journal on Optimization*, Vol. 16, No. 1, 2005, pp. 170–192.

[66] Zhai, Y. and Shah, M., "Visual Attention Detection in Video Sequences using Spatiotemporal Cues," *ACM International Conference on Multimedia*, 2006.

[67] Guo, C., Ma, Q., and Zhang, L., "Spatio-Temporal Saliency Detection using Phase Spectrum of Quaternion Fourier Transform," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2008.

[68] Achanta, R., Estrada, F., Wils, P., and Süsstrunk, S., "Salient Region Detection and Segmentation," *IEEE International Conference on Computer Vision*, 2008.

[69] Borji, A., Cheng, M., Jiang, H., and Li, J., "Salient Object Detection: A Benchmark," *IEEE Transactions on Image Processing*, Vol. 24, No. 12, 2015, pp. 5706–5723.

[70] Achanta, R., Hemami, S., Estrada, F., and Süsstrunk, "Frequency-Tuned Salient Region Detection," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 1597–1604.

[71] Achanta, R. and Süsstrunk, S., "Saliency Detection using Maximum Symmetric Surround," *IEEE International Conference on Image Processing*, Hong Kong, China, Sept 2010.

[72] Cheng, M., Zhang, G., Mitra, N., Huang, X., and Hu, S., "Global Contrast Based Salient Region Detection," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2011, pp. 409–416.

[73] Cheng, M., Warrell, J., Lin, W., Zheng, S., Vineet, V., and Crook, N., "Efficient Salient Region Detection with Soft Image Abstraction," *IEEE International Conference on Computer Vision*, 2013, pp. 1529–1536.

[74] Shi, J., Yan, Q., Xu, L., and Jia, J., "Hierarchical Image Saliency Detection on Extended CSSD," *IEEE Transactions on Pattern Analysis Machine Intelligence*, Vol. 38, No. 4, 2016, pp. 717–729.

[75] Fawcett, T., "An Introduction to ROC Analysis," *Pattern Recognition Letters*, Vol. 27, 2006, pp. 861–874.

[76] Meyer, F., "Color Image Segmentation," *IET International Conference on Image Processing and its Applications*, 1992.

[77] Rother, C., Kolmogorov, V., and Blake, A., "GrabCut-Interactive Foreground Extraction using Iterated Graph Cuts," *ACM Transactions on Graphics*, Vol. 23, No. 3, 2004, pp. 309–314.

[78] Lepetit, V., "EPnP: An Accurate O(n) Solution to the PnP Problem," *International Journal of Computer Vision*, Vol. 81, No. 2, 2009, pp. 155–166.

[79] Fischler, M. and Bolles, R., "Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography," *Graphics and Image Processing*, Vol. 24, No. 6, 1981, pp. 381–395.

[80] Shi, J., Ulrich, S., and Ruel, S., "Spacecraft Pose Estimation using a Monocular Camera," *Proceedings of the International Astronautical Congress*, Guadalajara, México, Sept 2016, IAC-16-C1.3.4.

[81] SSP-30219, "Space Station Reference Coordinate Systems," Jun 2005, NASA International Space Station Program, Rev H.

[82] Goodman, J., "Rendezvous and Proximity Operations of the Space Shuttle," Jul 2005, NASA Technical Reports Server.